

Master's Thesis

Data-driven Disability Persona
Augmentation for Service Journey
Inference

– Focusing on Daily Living Support Needs of People
with Developmental Disabilities –

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The Graduate School

Sungkyunkwan University

Department of Service Design Interdisciplinary Major

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ABSTRACT

Data-driven Disability Persona Augmentation for Service Journey Inference

Despite the clear necessity and potential value of service design for people with disabilities, it remains underrepresented in both academic and practical domains due to the dual limitations of population scarcity and low market viability. In particular, the extreme behavioral diversity among people with developmental disabilities—stemming from spectrum-based variability—makes generalization and representativeness especially difficult, posing fundamental challenges to quantitative analysis in service design and evaluation.

To address these structural issues—scarcity, spectrum diversity, and behavioral fragmentation—this study proposes the *Augmented Persona Mapping Framework*. The framework integrates data augmentation and algorithmic clustering to derive representative personas and maps them onto daily life scenarios to quantitatively assess support needs in daily living.

This research was designed around two core objectives. The first is to develop an algorithm that generates representative personas by augmenting a small set of real-world cases to reflect spectrum-level diversity. The second is to apply these personas to common daily living scenarios, structurally identifying activity-specific barriers and required functions, and visualizing support needs

through a quantitative analytical framework.

This study expanded interview data from six individuals with developmental disabilities into 2,406 synthetic cases using a K-Vineland-II - based data augmentation process. Principal component analysis and hierarchical clustering analysis were then used to extract five representative personas categorized by functional characteristics. These personas were applied to real-life scenarios—such as school routines, transportation use, meal preparation, and shopping—and large language models were employed to interpret both functional and adaptive behavioral limitations and quantify the required level of support. The results were visualized as structured journey map.

Validation was conducted by comparing the framework's outputs to real performance data from two people with developmental disabilities not included in the training data. The representative personas demonstrated over 75% alignment with actual disability levels, and the journey maps showed more than 86% similarity in activity patterns, confirming the framework's effectiveness as an analytical tool for identifying service intervention points.

This study provides empirical evidence for structuring and amplifying behavioral data for use in persona modeling and service need mapping. It introduces a replicable and data-driven alternative to traditional qualitative methods in disability-focused design. Furthermore, the proposed framework holds future potential for simulation-based service strategy development, particularly when integrated with digital twin environments and AI-driven service design.

**Keywords : Data-driven, Service Design, Developmental Disabilities,
Augmentation, Daily Living Support Services**

Chapter 1. Introduction

1.1 Background

1.1.1. From Functionality to Disability, and Service Design

Disability is no longer viewed solely as an individual issue, but rather as the cumulative result of complex interactions between a person's physical conditions and the social environment in which they live[1]. However, prevailing perceptions of disability continue to reflect a traditional medical approach, interpreting disability as a “problem” confined to the individual body. This perspective reduces the root cause of all disadvantages experienced by people with disabilities to their functional limitations, thereby placing the burden of resolving disability on the individuals themselves—whether by striving to restore normal functioning or by demonstrating competence despite their impairments. As a result, people with disabilities are often excluded, underestimated, pressured to conform to questionable norms, and perceived as inherently incapable[2, 3].

Over time, perspectives on disability have evolved. Although disability is often defined as a limitation in performing activities of daily living, impairments based on a medical model—such as structural damage to the body—do not necessarily constitute disability in functional terms. For instance, tasks such as lifting heavy objects, communicating in a foreign country, or delivering a speech in a loud voice only become disabling when a person is unable to carry them out. In this

sense, disability arises from a mismatch between a person's abilities and the demands of a specific context—factors such as aging, language, or health can all contribute to such misalignments[4]. Accordingly, the World Health Organization (WHO) regards disability as a universal aspect of the human experience. Every individual can experience varying levels of disability throughout their lives, meaning that disability should be understood as a spectrum that spans from full ability to significant impairment, rather than a binary condition of either presence or absence[1, 5]. Building on this view, the International Classification of Functioning, Disability and Health (ICF) proposes a bio-psycho-social model that integrates both the medical model—which focuses on the individual's health condition or impairment—and the social model, which emphasizes the environmental barriers that contribute to disability[1]. Within this framework, most of the elements that render an impairment disabling are not solely attributable to the body, but are instead shaped by the configuration of physical and social environments and the influence of institutional norms[2]. Therefore, a service design approach is required—one that explores and generates new forms of value relationships within socio-material configurations composed of diverse actors, including people, technologies, and objects[6].

1.1.2. Disability Design Approaches

Service design is a multidisciplinary field that contributes to service innovation by generating new ideas through design thinking, with a core focus on designing customer experiences[7, 8]. It involves designing the domains, scopes, and contexts in which interactions between users and services occur,

encompassing both tangible and intangible elements[9, 10]. As most goods and services now exist along a continuum from tangible-dominant to intangible-dominant, solutions related to disability are no exception to this trend[11]. Products, services, and systems that support the daily lives and enhance the quality of life for people with disabilities require technology-based assistance to improve accessibility, thereby demanding an integrated approach that combines product and service design. As technology advances, the scope and potential of such integrative design approaches continue to expand[12].

Approaches to improving the accessibility of products and services for people with disabilities are not limited to service design. Universal design aims for environmental inclusivity by designing products and environments to be usable, accessible, and understandable by all people without the need for adaptation or specialized design. It is applied across a range of public services, including architecture and urban planning, healthcare, and educational systems[13, 14]. Assistive technology, a subset of rehabilitation engineering, primarily targets short-term recovery or long-term functional support for individuals affected by injury or illness[15].

However, these existing approaches have been criticized for relying on supplier-centered perspectives rather than user-driven needs, often resulting in technology being confined to expensive, specialized products[12]. This tendency arises when third-party payers—such as health insurance providers or associations—are treated as the primary clients rather than the people with disabilities, or when there is a lack of deep understanding of end users' needs and lived experiences[15, 16]. In response, alternative perspectives argue for the necessity of personalization, given the diversity of experiences shaped by the

interaction between disability, individual characteristics, and environment[17]. Yet personalization typically requires advanced technology and entails significant costs, potentially placing additional economic burdens on users with disabilities[18]. Consequently, there has been a growing emphasis on inclusive design—an approach that, by embracing adaptability, seeks to design products and services in ways that account for user differences while ensuring broad accessibility and usability for as many people as possible. This principle aligns with strategic objectives for market expansion[19].

1.1.3. Market-oriented Approach to Disability through Service Design

Despite the overall size of the disability-related market, it remains fragmented into a wide array of smaller segments, each shaped by particular types of disabilities or activity limitations—such as motor, sensory, or cognitive disabilities[16]. Accordingly, people with disabilities are often treated as distinct customer segments with unique needs[20]. However, overly specialized or medicalized products tend to highlight disability as a core part of the user’s social identity, thereby posing a risk of stigmatization[16]. Moreover, assistive technologies designed to meet such specialized needs typically require high levels of technological sophistication and customization, making it difficult to achieve economies of scale. As a result, despite often having limited purchasing power, people with disabilities are faced with products that demand high costs[18, 21]. In this context, customer loyalty in the assistive technology sector tends to be driven by a perceived balance between cost and service, rather than satisfaction—grounded in the belief that a suboptimal product is preferable to the absence

of autonomy[16].

Such challenges can be addressed not by confining disability-related products to a specialized market, but by integrating the needs of people with disabilities into the design of mainstream products and services from the outset. This aligns with the core tenets of universal and inclusive design, which advocate for the incorporation of user diversity into mainstream design processes. Similarly, assistive technologies should not be positioned solely for specific user groups, but instead expanded in ways that also appeal to and benefit non-disabled users. Therefore, in order to sustain the growth of the disability market, a market-expansive approach that reflects the desires and expectations of all potential users—including those with disabilities—must be critically examined [15, 16].

To achieve this, a service design approach is essential—one capable of integrating diverse stakeholder perspectives and addressing complex challenges in a holistic manner. However, current applications of service design in the disability domain face significant limitations, including a lack of data and user scenarios that reflect real-life contexts of people with disabilities, limited direct participation of disabled users in the design process, and the absence of standardized design criteria[22]. These structural constraints underscore the need for a systematic investigation into service design methodologies specifically tailored for the disability context.

1.1.4. Data-driven Service design for Developmental Disabilities

As customer demands become more heterogeneous, the complexity and uncertainty of service design increase accordingly[23]. This presents a fundamental constraint in applying service design to the disability domain—particularly in the context of developmental disabilities. Developmental disabilities are spectrum disorders that cannot be easily defined by a single symptom or standardized pattern; functional differences among individuals vary widely depending on their developmental stage[24]. As a result, there are inherent limitations in quantitatively measuring or generalizing user needs, making it difficult to design products and services that comprehensively address the everyday lives or specific situations of people with developmental disabilities. From a market perspective, this complexity also poses challenges in formulating strategies that encompass a broad range of users.

However, these very conditions of uncertainty and diversity underscore the importance of data-driven approaches to service design. Such approaches offer the potential to quantitatively collect and analyze user experiences, enabling the structuring of individualized needs and their translation into reproducible design foundations[25]. Recent advancements in artificial intelligence, particularly large language models, have further enabled simulation-based methods to address user groups for whom real-world experimentation is ethically or practically constrained[26, 27]. For instance, based on limited behavioral data, virtual users with similar characteristics can be generated and placed into various scenarios to predict their behavioral responses. This allows for a transition from qualitative, interpretive analyses of user needs to more quantitative and

reproducible models.

Through this data-driven approach, the scattered and spectrum-like nature of disability-related issues—traditionally addressed in qualitative terms—can be reframed into quantifiable domains. By matching new user data with existing behavioral patterns, more efficient, scalable, and predictive service design becomes possible. Furthermore, simulation-based tools can anticipate user behaviors within specific contexts, enabling the preemptive diagnosis and resolution of real-world problems. Such methods reduce the temporal and economic costs associated with experimental design and service planning for both disabled and non-disabled users, thereby supporting the foundation for market-expansive design strategies.

1.2 Research Objective

The objective of this study is to propose a methodology for structurally analyzing problems, friction points, and service needs in the daily lives of people with developmental disabilities—whose spectrum nature makes qualitative approaches inherently difficult—by constructing representative personas through a data-driven approach.

To this end, the study pursues two practical tasks. First, it transforms the behavioral characteristics of people with developmental disabilities into structured data, thereby reconfiguring the highly diverse and scattered domain of developmental disabilities into a format suitable for data-based persona analysis. This includes developing an algorithm that augments limited real-world data into a larger dataset, addressing structural limitations in data collection. Second, it analyzes daily living data to define discrete units of support needs within the category of daily living, and matches them with the constructed personas to quantitatively identify and structure service demand.

Academically, this study demonstrates the potential for quantitative and automated approaches in user research on disability, thereby addressing the limitations of small sample sizes and interpretive subjectivity inherent in traditional qualitative methods. From a practical perspective, it offers strategic pathways for designing services and products for minority user groups often overlooked due to limited market potential. It also provides a foundation for inclusive service strategies that identify shared experiential touchpoints between disabled and non-disabled users, enabling market-expansive design strategies.

1.3 Outline

This thesis explores a data-driven service design process for analyzing daily living support needs among people with developmental disabilities, and is organized into five chapters. Chapter 1 introduces the background, objectives, and rationale of the study, and outlines the overall structure of the thesis. Chapter 2 presents the theoretical background, establishing a conceptual foundation by reviewing existing literature on disability models, service design, persona- and scenario-based journey mapping, developmental disabilities, and daily living support services. Chapter 3 proposes a data-driven service design framework for people with developmental disabilities. It presents a research model that augments small-scale disability datasets to generate representative personas and quantitatively analyzes service needs based on these personas. Chapter 4 applies the proposed model through an empirical study in two parts. Part 1 derives representative personas from real-world data and validates their representativeness. Part 2 constructs scenario-based journey maps for daily living tasks and analyzes each persona's capability to perform them, thereby evaluating service needs. Chapter 5 summarizes the findings, discusses the academic and practical implications of the proposed service design process, and concludes the study.

Chapter 2. Literature Review

2.1 Disability

2.1.1. Historical Evolution of Theoretical Models of Disability

The term *disability* has carried various meanings throughout history. In traditional views, disability was primarily defined based on impairments in bodily structure. However, such impairments do not necessarily result in limitations in performing daily tasks, and in many cases, they can be effectively compensated for. Accordingly, the conceptual focus shifted from bodily impairment itself to the resulting activity limitations. *Activity* refers to actions or tasks that an individual can or is expected to perform. In its 1980 publication of the ICIDH, the WHO defined disability not as the impairment itself, but as the limitation in performing activities that results from it.

However, the concept of *activity limitation* alone has limitations in fully capturing the functional status of an individual. Human abilities vary across a wide spectrum depending on the type of task, and the complexity or environmental demands of an activity can impose varying levels of limitation regardless of the presence or absence of disability. Consequently, the concept of disability has evolved toward understanding it as the inability to perform necessary activities within a given context. From this perspective, disability is redefined as a restriction in participation—a limitation in fulfilling social roles

due to a mismatch between the capabilities required to perform a task and the contextual demands of that task[4].

The *medical model* attributes the source of disability to individual impairment. It views disability as a problem that should be corrected or restored through treatment or rehabilitation, aiming to approximate a “normal” level of functioning. Under this model, people with disabilities are expected to adapt and learn to function despite their impairments. As a result, the responsibility for overcoming disability-related challenges is placed squarely on the individual.

In contrast, the *social model* understands disability not as a personal defect but as the result of environmental barriers. Some individuals with congenital disabilities do not perceive their condition as pathological, but rather accept it as a mode of existence. For instance, many people with visual or hearing impairments do not long for a non-disabled state, but instead regard their condition as part of their identity rather than a deficit. This perspective—viewing disability as a misalignment between the individual and their environment—has gained momentum through disability activism and sociological theory, contributing to a broader and more inclusive conceptualization of disability[2, 28].

Transcending the binary opposition between these two competing models, the WHO introduced the International Classification of Functioning, Disability and Health (ICF) in 2001. The ICF conceptualizes disability as the result of interactions between an individual’s health condition and their surrounding environment. It defines disability as a comprehensive concept encompassing impairments, activity limitations, and participation restrictions. In this model, disability is understood as a multidimensional construct, comprising bodily

impairments, limitations in capability, environmental factors, and restrictions in participation[1, 4, 29].

2.1.2. ICF: A Multidimensional Classification of Disability

The ICF is currently the most authoritative framework used worldwide for classifying disability across a range of purposes—including research, policy-making, education, clinical practice, and rehabilitation[1]. As illustrated in Figure 2-1, the ICF is structured around two primary dimensions: *Functioning & Disability* and *Contextual Factors*.

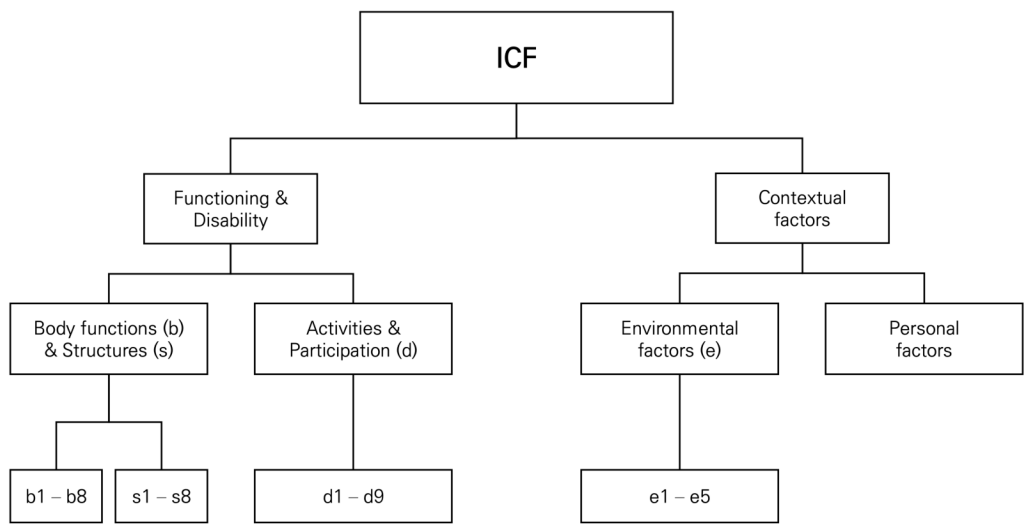


Figure 2-1. Structure of ICF

Note. Adapted from *Definition and assessment of disability in mental disorders under the perspective of the international classification of functioning disability and health (ICF)*, by Linden, M. (2017), *Behavioral Sciences & the Law*.

The Functioning & Disability domain is further divided into two components: *Body Functions & Structures* and *Activities & Participation*. Body functions refer to the physiological and psychological functions of body systems, including mental functions. Body structures are concerned with the anatomical parts of the body such as organs, limbs, and their components.

The Contextual Factors domain includes both *Environmental Factors* and *Personal Factors*. Environmental factors are subdivided into categories such as products and technology, the natural and built environment, social support and relationships, societal attitudes, and institutional services, systems, and policies. Personal factors refer to an individual's background characteristics, such as age, gender, and personality traits[4, 30]. Accordingly, a comprehensive understanding and evaluation of disability must consider not only bodily impairments but also a wide range of contextual conditions and life circumstances.

2.1.3. Established Strategic Approaches to Disability

Globally, over one billion people are classified as having some form of disability, accounting for approximately 16% of the world's population. Given this scale, a multifaceted approach to addressing disability-related challenges is essential. In particular, ensuring accessibility in interactive systems and services has emerged as a critical objective[22]. To achieve this, various design strategies have been traditionally employed, with universal design, assistive technology, and inclusive design serving as key methodologies for realizing accessibility.

Universal Design

Accessibility constitutes one of the core principles of universal design[22]. Universal design is a design strategy for achieving environmental inclusivity and is defined as: “the design of products and environments to be usable by all people, to the greatest extent possible, without the need for adaptation or specialized design”[13]. This concept was first introduced in 1998 by architect Ronald Mace at North Carolina State University and has since evolved based on a series of principles, as presented in Table 2-1[14].

Table 2-1. 7 Principles of Universal Design

Number	Principle	Definition
# 1	Equitable use	Designed to be usable by people with diverse abilities, ensuring both accessibility and market reach.
# 2	Flexibility in Use	Accommodates a wide range of individual preferences and abilities.
# 3	Simple and Intuitive Use	Easy to understand and use, regardless of the users' experience, knowledge, language skills, or focus.
# 4	Perceptible Information	Communicates necessary information effectively, regardless of environmental conditions or user's senses.
# 5	Tolerance for Error	Minimizes hazards and adverse consequences of accidental or unintended actions.
# 6	Low Physical Effort	Can be used efficiently and comfortably with minimum fatigue.
# 7	Size and Space for Approach and Use	Provides appropriate size and space for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

Note. Adapted from *Conceptual Challenges Between Universal Design and Disability in Relation to the Body, Impairment, and the Environment*, by Erkilic, M. (2011), *Disability and Health Journal*.

Built upon these principles, universal design has been applied across diverse fields including architecture and urban planning, product design, software and ICT, and education[31]. Although the concept emerged within disability discourse, universal design does not focus on specific groups with disabilities. Rather, it avoids specialized or adaptable solutions and instead aims to create universally usable forms that include all users. From this perspective, even the use of assistive technology is sometimes regarded as a form of “special design” and thus excluded. This tendency is rooted in the recognition that the diversity and complexity of disability experiences make it difficult to reflect all conditions fully in the design process or to reduce disability to a single identity. Accordingly, universal design adopts a practical strategy of “one design for all” to achieve inclusivity. However, this approach may result in outcomes that expect users to adapt to the design, effectively shifting the burden of accommodation onto the individual rather than the environment—ultimately reinforcing a provider-centered perspective[14].

Assistive Technology

Assistive technology encompasses a wide array of products, systems, and services aimed at compensating for loss of autonomy and supporting daily living and social participation for people with disabilities. It offers various benefits, including enhanced functionality, greater independence, improved safety, increased social inclusion, and expanded opportunities for education and employment. Although the assistive technology market may appear large, it is in fact fragmented into numerous niche markets based on functional impairments—such as motor, sensory, or cognitive disabilities. As a result, the market is

characterized by segmented structures, quasi-monopolistic competition, and a concentration of small-scale firms. However, due to the inherent difficulty of designing a single product that suits everyone, markets for customized solutions tailored to individuals with specialized needs continue to exist[16, 32].

Standardized, one-size-fits-all solutions are often insufficient to address the complex and individualized needs of people with disabilities. Accordingly, assistive technologies are increasingly personalized and adjusted to the specific contexts and requirements of each user[32]. Recent advancements in computer-based systems, including motion-recognition technologies like cameras and sensors, have enabled the development of more personalized and non-invasive assistive resources[33].

However, as the technological complexity of assistive devices increases, so too does their cost—creating a significant barrier to accessibility[34]. Consequently, the third parties, such as government agencies, health insurance providers, disability organizations, or charitable institutions, are often regarded as the actual purchasers of these devices. These third-party payers often hold significant decision-making power, leaving users with limited agency in the selection process and rendering them passive recipients rather than active consumers. As a result, assistive technology developers frequently regard these institutional payers—not the disabled users themselves—as their primary clients, conceptualizing users more as “patients” than “consumers.” This tendency reflects a broader lack of understanding of actual user experience and purchasing preferences[15, 16].

Inclusive Design

Inclusive design is an approach to designing mainstream products and services with the aim of enabling access and use by as many people as possible without the need for special or adapted design[35]. Because of this inclusive orientation, the concept is sometimes used interchangeably with universal design or “design for all,” though it is often interpreted as broader and more accommodating[36].

While both universal and inclusive design emphasize accessibility and inclusivity, inclusive design differs in its recognition that it may not always be feasible or appropriate to meet the needs of the entire population with a single solution. Instead, it assumes user diversity as a given and seeks to minimize exclusion by providing adaptability and choice across a range of user needs. Reflecting this pragmatic flexibility, definitions of inclusive design often include the qualifier “where reasonably possible”[19].

To realize these aims, inclusive design employs a variety of user-centered methodologies such as focus groups, video ethnography, surveys, critical user forums, workshops, empathic simulations, structured and unstructured interviews, and cultural probes. These methods ensure that user needs are maximally integrated into the design process[35, 36].

Ultimately, inclusive design aligns with the global trend toward integrating marginalized groups such as people with disabilities and older adults into the mainstream, rather than treating them as separate subpopulations. Strategically, this reflects a paradigmatic shift away from the medical model—which attributes disability to individual impairment—toward the social model, which sees disability as a consequence of poor design, exclusionary services and environments, and ingrained cultural stereotypes[37].

2.2 Service Design

2.2.1. Objective of Service Design

There are four major models that conceptualize the nature of “service” as the object of service design. First, from a marketing perspective, services are fundamentally understood as intangible. Because customers cannot directly experience the service prior to consumption, they rely on tangible evidence—such as advertisements, tickets, or service personnel—to perceive the quality of the service. This perspective led to the development of the service blueprinting technique, which distinguishes between visible and invisible components of service delivery. By visualizing internal production activities that exist below the line of visibility, this method emphasizes the importance of designing and coordinating tangible cues.

Second, the managerial perspective conceptualizes service through three core components: (1) the service outcome, which is the result perceived and valued by the customer; (2) the customer process, through which the customer actively participates; and (3) the service prerequisites, which are the resources provided by the service provider. Rather than delivering the outcome itself, the provider offers these prerequisites, which the customer uses to co-create the service outcome. The prerequisites consist of the service concept, service process, and service system, all of which are the primary focus of service design.

Third, from an engineering perspective, the service process is the central object of service design. It comprises a series of functionally organized activities,

including service operation activities carried out by the provider and customer service activities involving customer interaction. These processes are structured hierarchically, and designers must decompose and optimize them using engineering-based approaches to ensure effective management.

Fourth, in the economic perspective, services are viewed as a set of interacting characteristics: the service characteristics delivered to the customer, the technical characteristics that enable them, and the capabilities of both the provider and the customer. This model highlights the systemic nature of services, involving both tangible and intangible components, and emphasizes the mutual influence of provider and customer capabilities.

Based on this analysis, the scope of service design can be summarized into three key domains: (1) exchange relations between service providers and customers, (2) interfaces and infrastructures that facilitate interaction, and (3) the materiality through which services are embodied[11].

Thus, services are complex systems that integrate intangibility and tangibility, producer-consumer interactions, and technological and organizational infrastructures. Service design seeks to understand these elements holistically and reconfigure them from a user-centered perspective to innovate the overall service experience. It functions not merely as a tool for designing touchpoints, but as a strategic design approach that reorganizes entire service systems to create sustainable value[38].

2.2.2. Framework of Service Design

Service design is a multidisciplinary field that contributes to service innovation by delivering new and meaningful experiences to users through design thinking[7]. As societies continue to transition from product-based to service-based economies, there is growing recognition that value is now co-created through ongoing interactions between companies and customers. This shift is driven by factors such as increasingly diverse customer needs, the dematerialization of many products, and new production methods. Service design, which once focused on improving discrete customer touchpoints, now aspires to redesign multi-actor service systems for deeper organizational and social outcomes. In practical terms, this means that organizations must not only adopt user-centered mindsets but also develop collaborative capabilities to address complex societal challenges in fields ranging from healthcare to education[39]. Accordingly, service design has evolved into a multidisciplinary domain known as “service science,” integrating perspectives from design, business, engineering, and the social sciences. In the course of this development, temporary ambiguity and fluidity in its definitions, methodologies, and terminology have emerged. However, growing academic discourse is gradually establishing shared conceptual understandings and frameworks[40].

Service design is a human-centered and iterative design approach aimed at creating or improving services in ways that generate customer value. It follows a cyclical process of “design - validate - measure - redesign,” in which problems are redefined, information is gathered and interpreted, ideas are generated, and solutions are developed and evaluated. In particular, the human-centered

approach is advantageous for identifying unexpected insights and producing precise, innovative solutions grounded in a deep understanding of the user's context[7, 38].

At its core, service design involves integrating and aligning the perspectives of multiple stakeholders through the use of design methods, design research, design thinking, and visualization techniques. Various process models have been proposed by scholars and institutions, ranging from three to five stages. As of now, there is no standardized model for the service design process[38].

One of the most widely adopted frameworks is the five-stage model based on design thinking: Empathize - Define - Ideate - Prototype - Test. This model originated from the networks of IDEO, d.school, and the Hasso Plattner Institute[41]. *Empathize* refers to understanding the user's senses, emotions, thoughts, motivations, and personality. This goes beyond surface-level data collection and emphasizes observing and immersing in the user's life to discover latent needs and design opportunities. The qualitative data gathered at this stage is used to construct personas, enabling a multi-dimensional representation of user behaviors, contexts, and emotional flows. Methods commonly used include ethnographic observation, in-depth interviews, customer journey mapping, and co-creation. Group-level behavioral patterns may also be examined. *Define* involves organizing the data gathered in the Empathize stage and extracting the core problem. This step synthesizes fragmented information into patterns and insights, articulating a concrete design challenge. Rather than merely stating the problem, this stage requires strategically reframing it to guide the ideation process. A well-defined problem provides the foundation for the quality and direction of subsequent idea generation. *Ideation* is the stage of divergent

thinking, where a wide range of potential solutions are explored in response to the defined problem or user needs. This is followed by convergent thinking, which refines ideas based on feasibility and effectiveness. Multidisciplinary collaboration plays a crucial role in combining diverse perspectives and complementary knowledge for successful idea development. Common tools include brainstorming, visualization, mind mapping, diagrams, sketches, sticky notes, and whiteboard drawings. Storytelling techniques, narratives, and role-playing are also often employed. *Prototype* is the stage where selected ideas are visualized and materialized into low-fidelity mock-ups. In design thinking, prototypes are not polished final products but experimental tools that enable testing with minimal resources. The purpose is not merely to create a model but to engage users early, evaluate feasibility, and iteratively refine ideas. This approach aligns with the practical mantra: “Build small, fail fast, improve continuously.” *Testing* involves validating the developed prototype with real users and stakeholders. This stage goes beyond evaluation by repeatedly experimenting and collecting feedback to assess the solution in terms of desirability, feasibility, and viability. Testing is framed as a learning process—learning in action—in which feedback loops feed into earlier stages to improve and evolve the design. Common methods include field experiments, user feedback interviews, and attribute evaluation tests[42, 43].

Meanwhile, the Double Diamond model proposed by the UK’s Design Council has become a widely recognized framework that visualizes this process through two iterative cycles of divergence and convergence. The model consists of four stages: Discover - Define - Develop - Deliver, which align structurally with the five stages of design thinking[44]. The Double Diamond model has been

adapted flexibly in various contexts. Modifications include expanding or collapsing the scope of each stage, repeating or layering the diamonds, or adding a third diamond to form a Triple Diamond process. Some versions also introduce a front-end exploratory phase focused on user experience, or a post-implementation phase concerned with evaluation and sustainability. Despite structural variations, these approaches all share the core principle of the Double Diamond: iterative divergence and convergence for both problem finding and problem solving. In this sense, the model does not treat problems as fixed targets but seeks to redefine them within user contexts and resolve them through iterative experimentation. Thus, the Double Diamond functions not as a rigid sequence but as a flexible design-thinking framework that responds to the complexity and multi-layered nature of service design[45].

2.2.3. Data-driven Service Design

With the advancement of digital technologies, an unprecedented volume of data is being generated at a rapid pace from sensors, connected devices, and online platforms[46]. Customer behavior data is now collected in real time across various industries such as transportation, healthcare, hospitality, and logistics, providing a foundation for understanding user experiences with greater precision[47]. Developments in data collection and processing have accelerated efforts to integrate data and analytics as a unified unit of analysis in service innovation—including service design. This integrated approach is referred to as data-driven service design, and it has gained increasing attention in both academic research and industry practice in recent years[48].

Data-driven service design is typically applied in three key ways. First, in the *explorative* phase, data is analyzed to uncover previously unrecognized patterns, new opportunities, or actionable insights. Second, in the *validation* phase, data is used to empirically test generated ideas, hypotheses, and performance outcomes at different stages. Third, in the *generative* phase, data serves as the foundation for designing and delivering the final service or product-service system[25].

A primary contribution of data-driven service design lies in its enhancement of quantitative rigor and scientific validity in a domain traditionally rooted in qualitative methodologies. For instance, customer experiences and emotional responses can be quantified through hashtag and text analysis on social media; brand experiences can be structurally mapped via network analysis; and customer interactions at each touchpoint can be measured using behavioral data. Additionally, techniques such as process mining enable the visualization and analysis of customer pathways and actions. Based on such data, it is now possible to generate customer journey maps and service blueprints grounded in empirical evidence[49 - 55].

This approach not only reduces the time, cost, and effort required for the iterative cycles of divergent and convergent thinking in service design, but also facilitates the systematic structuring of complex user experiences[56]. It aligns with theoretical efforts to extend or modify the traditional Double Diamond model, offering new possibilities for data-integrated diamond frameworks. For example, Chae (2023) proposed the DDAD (Diamond - Diamond - Agile - Diamond) model, which utilizes social data and digital ethnography to design home appliance service scenarios. This model analyzes customers' lifestyles through data, derives data-driven personas, identifies relevant brand matches,

applies agile methodologies for simulation, and ultimately develops service scenarios based on the resulting insights[57]. Park (2023) introduced the 1H1D (One Hexagon One Diamond) model, which combines media literacy with service design to support government policy communication education. This model enhances participant understanding through the exploration of policy-related media content, followed by deep learning and data visualization activities[56]. Lee (2025) proposed a Triple Diamond structure that extends the traditional Double Diamond process to support OSMU (One Source Multi Use) strategies in the webtoon industry. This model adds an exploratory data insight phase that precedes the conventional design process, enabling the identification of meaningful patterns and opportunities from multiple sources, such as market data, user behaviors, and content[58]. These examples illustrate how data-driven service design enables a quantitative approach to service innovation across diverse domains. At the same time, they highlight a critical limitation: the contextual and emotional dimensions of user experience remain difficult to fully capture with data alone. Thus, future directions in service design must explore how to quantify and integrate qualitative insights, pursuing a balanced and hybrid approach that enables more practical, iterative, and replicable design frameworks.

2.2.4. Disability Service Design

The inherent characteristics of service design, which consider the perspectives of various stakeholders, can be effectively applied to disability-related issues. Services for people with disabilities often involve complex design problems that

require the coordination of diverse needs and understandings from service providers, designers, third parties such as governments, and actual users including people with disabilities and their families. Service design is particularly useful in such multilayered stakeholder environments because it takes into account both user contexts and environmental interactions[20].

Services for people with disabilities are actively studied across a wide range of domains, including smart homes, healthcare, and education, with a focus on personalized design and improved accessibility. However, the effectiveness and scalability of disability service design are often hindered by several limitations: lack of real-world data on disabled individuals, insufficient direct participation from users with disabilities, absence of standardized design guidelines, and insufficient consideration of social demands and physical or environmental contexts, along with a lack of concrete use-case scenarios[22]. In fact, major international accessibility standards, such as the ADA and IEC63008, are often limited to specific types of disabilities and provide insufficient rationale or application-ready scenarios to be effectively implemented in real-world product and service design[59].

To address these issues, a growing number of studies have begun to adopt concepts and methods from service design in the disability domain. Lee et al. (2021) constructed personas that reflect the characteristics of various disabilities – such as visual, auditory, and mobility impairments, as well as aging – through in-depth interviews and observations, and identified specific accessibility issues and needs in real-world contexts[59]. Burke & Heller (2017) conducted surveys to measure unmet service needs among people with developmental disabilities and explored relevant factors[60]. Lim et al. (2017) developed a Context-Based

Activity Modelling methodology that integrates contextual information and activity flows for customized service design for people with disabilities. This was applied to the case of a kneeling bus to verify its utility in PSS design processes[17].

Accessibility is a central consideration in disability service design, and as the importance and utility of accessible services become increasingly emphasized, service providers are expected to incorporate this actively. However, due to limited social attitudes and responsiveness among service providers, it remains difficult to meet the service expectations of people with disabilities. Institutional, informational, and attitudinal barriers reduce service accessibility and lead to broader limitations in everyday life. These issues stem from the fact that disability services have traditionally been designed based on a medical model rather than a social one. Furthermore, for service providers, designing for accessibility often falls behind other priorities due to financial constraints, the diverse and complex needs of disabled users, and the lack of standardized frameworks[20].

To solve these problems, it is essential to reposition accessibility not merely as a responsibility or cost but as a market-driven priority. If accessible service design can be aligned with profitability, it will naturally become a strategic concern for companies and institutions[61]. This view is supported by the idea that the vulnerability of disabled consumers does not solely result from their impairments, but rather emerges from the interaction between disability and markets or policy regulations designed around non-disabled norms[62]. Therefore, it is crucial not only to develop assistive technologies and special services exclusively for people with disabilities but also to pursue inclusive and

universally accessible products, services, and systems. In fact, technologies like touchscreens and voice recognition originated as assistive technologies but have since expanded into the mainstream. This demonstrates that insights and technologies derived from assistive applications can enhance overall user experience, underscoring the importance of this trajectory[16].

Nevertheless, disability-oriented design continues to face numerous structural and technological challenges, such as the need for high-level design approaches that integrate diverse demands, difficulties in acquiring realistic user data, and challenges in simulating real-life usage contexts[22]. Therefore, it is necessary to develop a more multidimensional approach to service design—one that is grounded in a deeper understanding of people with disabilities and enables the quantification and targeted response to their service needs.

2.3 Persona

2.3.1. Typology, Composition, and Construction Methods of Persona

A persona is a realistic, fictional character that represents the characteristics, needs, behaviors, goals, and expectations of target users, synthesized from multiple individuals who share similar motivations, goals, and behavioral patterns[59, 63]. Designers often incorporate personas to ensure that insights from numerical data analyses and more descriptive, user-focused studies are both adequately addressed[64]. Typically, one to seven personas are developed per project, and their differentiation is based more on goals and behaviors than on demographic characteristics[65].

According to various studies, the specific definition of a persona differs depending on the applied methodology, as shown in Table 2-2, and is shaped by the strategies employed by UX professionals to understand the perspectives of key users[26, 66, 67].

Table 2-2. Type of Persona

Type	Explanation
Goal-oriented	The most basic and traditional approach, where user groups are segmented based on shared goals and objectives to derive archetypal user types.
Role-based	Focuses on the user's roles and life cycle, emphasizing behaviors connected to the user's goals and responsibilities.
Ad-hoc	A persona derived from Norman's approach, constructed without extensive user research and based primarily on the designer's intuition and experience.
Engaging	Defined by Nielsen, this approach assumes that user needs arise from individual traits and are expressed in specific contextual scenarios. It emphasizes vivid, narrative-driven personas to foster empathy, insight, and understanding.
Holistic	Integrates five key dimensions—personality, intelligence, knowledge, cognition, and factual information—into a multidimensional representation to enable a comprehensive and in-depth understanding of the user.

Note. Adapted from *ChatGPT for learning HCI Techniques: A case study on interviews for personas*, by Barambones, J., Moral, C., de Antonio, A., Imbert, R., Martínez-Normand, L., & Villalba-Mora, E. (2024), *IEEE Transactions on Learning Technologies*; and *Persona ontology for user centred design professionals*, by Anvari, F., & Tran, H. M. T. (2013), in *Proceedings of the ICIME 4th International Conference on Information Management and Evaluation*.

However, due to the absence of a universally accepted template and the diverse layouts employed across different studies, both researchers and practitioners often face uncertainty about what information to include when constructing a new persona[68]. In practice, the composition of personas varies widely across research and design projects. According to Cooper, a persona typically consists of a name, a photograph or illustration, and a brief narrative description[64]. Nielsen et al. (2015) analyzed twelve persona templates used between 2006 and 2013, categorizing the information into two main groups: personality-related attributes (such as demographic and personality traits), and domain-specific design information (including technology use, daily routines, product-related goals, and behavioral patterns)[69]. Salminen et al. (2020) examined thirty articles and analyzed the graphical layouts of personas, as summarized in Table 2-3, classifying them based on levels of informational richness[68]. The simplest forms, considered persona skeletons, include only sparse and minimal data. Medium-level personas—typically the most common type—contain demographic and psychological attributes, along with daily life scenarios and user goals. Highly detailed personas provide richer and more comprehensive information directly relevant to their design purpose, often incorporating narrative elements to imbue the persona with lifelike qualities[68].

Table 2-3. Composition of Persona

Category	Containing information
Name	Full name, first name, or epithet
Age	Age (or age range) ascribed to the persona
Gender	Gender ascribed to the persona
Personality and psychographics	Character traits and disposition of the persona
Lifestyle	Living situation, leisure, work-life balance
Experience	The person's experience with the product
Daily work context	The persona's role and duties in the workplace
Product related behaviors	How the persona interacts with technology and/or tools in the workplace
Product goals	What the persona hopes to achieve
Scenarios	Specific events involving the persona in relation to the product
A day in the life	Daily context for persona in relation to the product
Market size	Sample size of analyzed population that matches a particular persona
Color-coding to indicate segment	Color tagging for details of the persona
Use of facial picture	Photograph of real person included
Use of cartoon picture	Cartoon image to represent persona
Reference to sources	Source of data or explanation of metrics
Disabilities	Handicaps of the persona (particularly for studies written in healthcare contexts)
International considerations	Cultural heritage, ethnicity and/or citizenship
International considerations Explanations	Tooltip definitions

Note. Adapted from *A template for data-driven personas: Analyzing 31 quantitatively oriented persona profiles*, by Salminen, J., Guan, K., Nielsen, L., Jung, S. G., & Jansen, B. J. (2020), in *Human Interface and the Management of Information. Designing Information: Thematic Area, HIMI 2020, Held as Part of the 22nd International Conference, HCII 2020, Copenhagen, Denmark, July 19 - 24, 2020, Proceedings, Part I 22* (pp. 125-144), Springer International Publishing.

The degree of complexity and descriptive richness found in a persona is largely a reflection of the research strategies utilized during its formation. Traditionally, personas have been developed using qualitative methods such as ethnographic studies, interviews, observations, and focus group interviews[70]. These qualitative personas have been widely used for their ability to provide rich contextual information about user behaviors[71]. However, synthesizing qualitative data requires manual analysis, which can be resource-intensive and reliant on subjective interpretation, thus raising concerns about objectivity and methodological rigor. Additionally, such approaches are often difficult to scale and are limited in terms of representativeness and generalizability[72, 73]. In response to these limitations, quantitative personas have been proposed as an alternative. Quantitative personas represent user groups in statistically valid ways, allow for algorithmic replication, and can be empirically validated using statistical indicators[68]. This approach utilizes demographic and behavioral data –such as survey responses and system log data–to identify trends, and applies multivariate analysis techniques such as factor analysis, principal component analysis, and cluster analysis to classify homogeneous user segments. Because they are grounded in large-scale user data, quantitative personas are often more representative and reproducible. However, they lack contextual depth and may pose interpretive challenges in understanding the behavioral meanings behind identified clusters or variables[71]. To overcome the limitations of both approaches, mixed-method personas have emerged, combining the strengths of qualitative and quantitative techniques. This hybrid approach enables a more comprehensive and balanced persona construction. For instance, survey data may be used to gain initial user insights, which are then refined through user

interviews and structured using principal component analysis and iterative clustering methods[74, 75]. Jansen et al. (2022) systematically compared the strengths and limitations of qualitative, quantitative, and mixed-method persona development. Table 2-4 reconstructs these findings[74].

Table 2-4. Persona creation method

Method	Explanation	Strength & Weakness	
Qualitative	Involves in-depth analysis of user motivations and behavioral contexts using small-scale qualitative data, such as FGIs and interviews.	Strength	- Enables deep analysis of complex and multi-layered user behaviors and contexts. - Effective for interpreting internal factors such as emotions and attitudes in individual cases.
		Weakness	- Relies heavily on subjective interpretation and manual analysis, leading to high cost and time consumption. - Limited scalability and generalizability for large-scale applications.
Quantitative	Uses statistical analysis of structured data collected via surveys, logs, APIs, etc., to identify behavioral patterns across user groups.	Strength	- Ensures statistical validity and allows for repeatable persona generation. - Capable of representing large-scale user groups with high efficiency. - Supports fast prediction and clustering, with potential for automation.
		Weakness	- Provides limited understanding of contextual depth; interpretation of clusters may be unintuitive. - Less effective at capturing complex behaviors, emotions, or underlying motivations.
Mixed-methods	Integrates qualitative and quantitative data collection and analysis to complement each approach.	Strength	- Balances statistical representativeness with contextual richness. - Enhances reliability and interpretability through multi-source data integration.
		Weakness	- Time- and resource-intensive due to the complexity of data collection and integration. - Requires expert interpretation due to the complexity of analysis and outcomes.

Note. Adapted from *How to create personas*, by Jansen, B. J., Jung, S. G., Nielsen, L., Guan, K. W., & Salminen, J. (2022), *Pacific Asia Journal of the Association for Information Systems*.

2.3.2. Data-driven persona

Recent studies have introduced a variety of terms referring to personas generated using computer-based tools, differentiated by the nature of the data employed or the method of generation. These include quantitative personas, data-driven personas, algorithmically generated personas, LLM-based personas, and AI-generated personas, as summarized in Table 2-5. Among them, the term *data-driven persona* broadly refers to personas constructed using real-world data related to users or customers, and thus, the other terms can be understood as subcategories of this overarching concept. Data-driven personas are not limited to quantitative data; qualitative data can also serve as a foundational input in their construction. However, in practice, the term tends to be applied predominantly within quantitative research contexts. The distinctions among these terms typically stem from the mode of data processing or the persona-generating entity (e.g., algorithm, large language model, AI system)[76].

Table 2-5. Subtypes of Data-driven persona

Method	Explanation
Data-driven persona	Generated based on qualitative and quantitative data about users or customers, typically using statistical techniques or machine learning. Offers high objectivity and accuracy but may lack contextual understanding due to dependency on data quality.
Quantitative persona	Constructed from structured quantitative data such as surveys, providing numerical insights and ease of data collection and analysis. However, it may be limited in capturing complex behaviors.
Algorithmically generated persona	Automatically generated through algorithms such as clustering. While efficient and objective, it often lacks interpretability regarding user characteristics and behavioral drivers.
LLM-generated persona	Generated using large language models, which can analyze large volumes of text data to extract deep insights. However, expertise in natural language processing is often required.
AI-generated persona	Created using general AI methods (e.g., clustering, rule-based systems), integrating multiple data sources to describe multifaceted user attributes. Interpretation can be difficult, and outcomes may lack transparency.

Note. Adapted from *PersonaCraft: Leveraging language models for data-driven persona development*, by Jung, S. G., Salminen, J., Aldous, K. K., & Jansen, B. J. (2025), *International Journal of Human-Computer Studies*.

A variety of data types and methodological approaches are used in the construction of data-driven personas. According to Joni et al.'s review of prior studies published between 2005 and 2020, the most commonly used data sources for data-driven persona research include surveys, web and social media data, personal big data, and customer behavioral data. Web and social media data encompass digital behavioral traces from platforms and user clickstream logs. Personal big data refers to sensor-based information collected from individual devices—such as GPS signals and biometric data—and is primarily used in health and wellness domains. Customer behavioral data includes real user interaction logs[77]. Although these quantitative data types are readily accessible through demographic surveys and market segmentation research, qualitative data—such as interviews, FGIs, and open-ended surveys—are also widely used in data-driven persona development. In practice, developers often construct initial personas based on quantitative data and refine them using qualitative insights to enhance contextual richness. More recently, digital interaction texts such as online chat logs have also been recognized as valuable qualitative sources[78].

The core methodology for constructing data-driven personas involves identifying user segments through clustering analysis. This analytical process may incorporate exploratory factor analysis (EFA), factor analysis (FA), principal component analysis (PCA), latent semantic analysis (LSA), multidimensional scaling, hierarchical clustering, and K-means clustering[77, 79]. Among these, hierarchical and K-means clustering are considered forms of cluster analysis (CA). PCA, FA, and CA are all powerful statistical techniques used in processing large-scale datasets by reducing dimensionality and minimizing computational complexity[78, 80, 81].

PCA was the technique originally used by Sinha (2003) in proposing a semi-automated quantitative method for persona clustering. As one of the earliest multivariate techniques, PCA extracts principal components—new variables derived from linear combinations of the original variables—to reduce dimensionality while preserving core informational structures. Sinha applied PCA to 32 attributes related to restaurant experiences, identified five principal components, and initially derived four personas. Based on further analysis of inter-factor distinctions, three final personas were proposed[82, 83].

FA, on the other hand, is a statistical method that assumes all observed variables are correlated through latent factors. It identifies the underlying structure among variables and eliminates irrelevant or redundant factors, enabling the selection of meaningful variables. Factor analysis includes two primary approaches: exploratory factor analysis (EFA) and confirmatory factor analysis (CFA). EFA is typically employed in the early stages of research when there is little prior knowledge about the underlying structure or latent factors among variables, allowing researchers to explore inter-variable relationships. In contrast, CFA is used to test whether a hypothesized factor structure—based on a specific theory or model—fits the observed data[84]. While PCA and FA are not theoretically identical, they tend to perform similarly under the pervasiveness assumption—that is, when common factors exert a strong influence across many variables in high-dimensional data[85]. Both methods serve as foundational tools in high-dimensional data analysis and, according to empirical findings, exhibit the highest consistency with expert-led manual clustering results[81].

While FA and PCA simplify data by identifying latent structures or underlying factors and reducing variables accordingly, cluster analysis simplifies

multivariate data by dividing it into a predefined number of clusters[78]. Each resulting cluster is constructed to maintain high internal homogeneity while maximizing heterogeneity between clusters[86]. Hierarchical clustering forms clusters based on the distance or similarity between data points. It operates by progressively merging or splitting items, starting from the most similar, resulting in a tree-like structure called a dendrogram. This dendrogram visually represents the hierarchical relationships among data points and allows for flexible determination of the number of clusters by cutting the tree at different levels. Several linkage methods are available depending on how inter-cluster distances are calculated, including single linkage (minimum distance), complete linkage (maximum distance), average linkage, and Ward's method (minimum variance)[87]. K-means clustering, like hierarchical clustering, is also based on distance or similarity between data points, but it is a non-hierarchical, centroid-based algorithm. The number of clusters, k , must be predefined by the analyst, and the result is sensitive to the choice of this value. The algorithm begins by assigning data points to the nearest initial centroids and then iteratively recalculates the centroids of each cluster. This process continues until the centroids converge and all data points are stably assigned to k clusters. Since the choice of k significantly affects clustering outcomes, techniques such as the elbow method and silhouette analysis are commonly used to determine the optimal number of clusters[80].

Unlike the previously mentioned methods, which rely on structured quantitative data, Latent Semantic Analysis (LSA) is designed for large-scale text data, such as interview transcripts. LSA applies statistical computations to extract and quantify the contextual meanings of words using specialized

software. It assesses word similarity based on their presence or absence in similar contexts, and its results often approximate human-level judgment in tasks such as standardized test scoring or essay evaluation[81].

These various data-driven persona generation techniques have been developed as alternatives to traditional qualitative personas, which often require extensive time and effort for manual clustering. While such methods enable the processing of large-scale quantitative data, they still rely on manual decisions for hyperparameter settings—such as the number of personas to generate and the scope of information to include. In response, recent research has turned to AI and large language models to further automate these decision-making elements in persona generation[76].

Table 2-6. Clustering methods for Data-driven persona creation

Method	Explanation
Principal Component Analysis (PCA)	A dimensionality reduction technique that derives principal components based on inter-variable correlations, summarizing key factors while minimizing information loss.
Factor Analysis (FA)	Simplifies variable structures by estimating shared latent factors among variables; can be applied in either exploratory (EFA) or confirmatory (CFA) form.
Hierarchical clustering	Forms a tree-structured dendrogram by progressively merging or splitting items based on similarity between data points.
K-means clustering	Iteratively partitions data into k predefined clusters using centroid-based distance calculations until convergence.
Latent Semantic Analysis (LSA)	Quantifies contextual meaning by analyzing word co-occurrence patterns in large-scale text data, identifying semantic similarities to uncover latent data structures.

2.3.3. AI-driven persona

Generative AI models such as OpenAI’s ChatGPT, Anthropic’s Claude, and Google’s Gemini integrate user information from vast data sources, including the internet and social media platforms[88]. The advancement of large language models (LLMs) has opened new possibilities for persona generation in various fields, including software engineering, UI design, and education. In particular, generative AI can be embodied as anthropomorphized agents capable of assuming specific roles. Despite some degree of bias in age, gender, or personality, these AI-generated personas can exhibit sufficient diversity, as well as internal consistency, empathy, and trustworthiness[89]. However, the field remains in its early stages, and fully autonomous persona generation systems are still lacking. As a result, various experimental approaches are currently being explored[90]. Unlike conventional automated persona generation, which has primarily focused on numerical data, LLM-based methods are data-agnostic and capable of analyzing and interpreting both quantitative and text-based qualitative data[76]. This ability to process textual data significantly broadens the scope of information that can be embedded in a persona—such as users’ expressed concerns, opinions, needs, and desires extracted from online posts and social content[91].

Traditional data-driven persona generation has evolved beyond mere analysis to include automated data collection through APIs, thereby constituting an early form of automated persona creation[92]. In contrast, AI-driven personas represent a more advanced and holistic development in this trajectory. Data-driven personas were originally developed to overcome the limitations of

manually constructed personas based on qualitative data. Yet, key processes such as data labeling and variable selection still required human intervention. In contrast, persona generation using LLMs allows for automation and intelligent participation across the entire process—including data labeling, semantic interpretation, and narrative construction—enabling a more comprehensive and autonomous persona creation framework. Studies have shown that AI-generated personas do not significantly differ in quality from those created by human designers[26, 76].

AI-driven persona generation is being explored through various approaches, and its applications are steadily expanding. First, some studies have generated personas through simple prompt-based approaches. For instance, Kocaballi (2023) utilized an LLM to create five hypothetical user personas and conducted simulated interviews with them. This method offers high accessibility, as it enables persona generation without the need for pre-existing data sets[93]. Other studies focus on automating traditional persona creation workflows based on human-generated data. Paoli (2023) employed an LLM to code and thematically analyze semi-structured interview transcripts, subsequently generating user personas from the derived insights. This demonstrates the potential for automating conventional qualitative analysis processes[94]. In a different direction, Jung et al. (2025) proposed an automated persona generation pipeline based on survey data. They developed a six-step procedure using an LLM to explore the feasibility of persona generation using quantitative inputs[76]. Barambones et al. (2024) took a hybrid approach by designing prompts that instructed the LLM to role-play users corresponding to pre-defined, goal-oriented personas derived from actual users. The resulting AI-generated

interviews were evaluated by UX practitioners and HCI experts. The findings confirmed that, with well-crafted prompts and configurations, AI-generated personas effectively reflected real user profiles, and the quality of the simulated interview content was comparable to that of human-generated data[26].

On another front, the expansion of AI-driven personas also suggests the potential to simulate real-world attitudes and behaviors in virtual environments. Park et al. (2024) trained AI agents using interview data from 1,052 demographically representative individuals in the United States, taking into account age, gender, race, region, education, and political ideology. These interviews covered various aspects of life, including childhood, education, family and relationships, and major life events. The resulting *Generative Agents* replicated the attitudes and behavioral tendencies of real individuals, enabling the simulation of their decision-making processes. In subsequent evaluations, these generative agents achieved approximately 85% alignment with actual individuals in responses to the GSS survey, the Big Five personality test, and standard behavioral economics experiments such as the dictator game, trust games (as both player 1 and 2), public goods game, and prisoner’s dilemma[95]. Ge et al. (2024) employed an LLM to infer user personas based on massive web-based textual data. Rather than generating isolated personas, their approach enabled the derivation of additional, emergent personas from the relationships among the initially generated ones. As a result, they produced over one billion personas, organized within a *persona hub* structure. This hub was conceptualized not merely as a collection of user models, but as a set of “distributed carriers of world knowledge”—a functional system for representing collective human understanding. Each persona embodied a distinct viewpoint and experience,

enabling the generation of responses to prompts through the synthesis of multiple perspectives rather than a single answer[96]. Altera developed between 500 and 1,000 AI agents using LLMs, assigning each agent a unique role and personality. These agents were designed to maintain consistent personality traits while responding flexibly to external stimuli. Within a Minecraft-based virtual environment, and without predefined scripts, the agents engaged in self-directed social interactions, gradually forming complex relational networks involving social norms, role distribution, cooperation, and conflict. This experiment demonstrated the potential of AI-driven personas to go beyond user modeling, toward simulating human-like life patterns, behavioral characteristics, and dynamic social systems[27].

2.3.4. Disability persona

Although AI-driven personas can be applied across diverse contexts, special caution is required when modeling marginalized users, such as people with disabilities. Since LLMs are trained on vast datasets from the internet, digital publications, news media, and social networks, they can inadvertently internalize societal biases against minority and underrepresented groups. Prior research has shown that LLMs can inadvertently reflect social biases connected to demographic factors like disability, ethnicity, age, and gender. These biases often manifest not through explicit hate but through the subtle reproduction of social stereotypes. Accordingly, creating disability personas necessitates a conservative approach, including the use of disability-sensitive training data and harm annotation frameworks to detect bias[97].

Despite limited research in this area, disability personas serve as a valuable and efficient means of incorporating the needs of people with disabilities into accessibility-focused projects, while minimizing the resource burden of direct participation. Recent EU research initiatives have actively employed disability personas[98]. For example, the *Designing with People* resource, developed by the Helen Hamlyn Centre for Design, introduces ten example profiles depicting people with disabilities. These case studies contain a range of background details, such as age, health status, daily routines, use of assistive devices, and personal priorities to support designers' understanding. However, because the platform does not specify how many people share these characteristics, its value for drawing generalizable conclusions remains limited[99].

Historically, personas for accessibility-sensitive users have typically involved

adding medical impairment data to existing persona structures. This has led to static, homogenized representations that fail to capture the complexity and diversity within the disability community. For instance, while deaf individuals are often socially categorized as disabled, many do not perceive themselves as such. This disconnect underscores the complex nature of identity, the diversity of ability-based experiences, and the contextual nuances that have often been overlooked or insufficiently addressed in the design of disability personas[28]. In addition, while various standards and guidelines exist for designing accessible products, they often lack specificity or cover only limited user groups. Therefore, there is a growing need for more refined persona models that reflect nuanced contextual factors, user characteristics, behavioral patterns, task barriers, goals, needs, pain points, and accessibility challenges[59].

The importance of disability personas lies in their potential to shift perceptions of people with disabilities from being seen merely as “exceptional cases” to being regarded as “lead users” in product and service innovation. *Lead users* are individuals who stand at the forefront of market trends, encounter emerging needs before the general user base, and are likely to derive significant advantages from the resolution of those needs through innovation. In this regard, people with disabilities—who often interact with products and environments under extreme or constrained conditions—can offer forward-looking insights for enhancing general user experience, forecasting emerging trends, and promoting inclusive design practices[100].

However, lead users are by nature a minority, and people with disabilities constitute a highly heterogeneous group. Even within the same disability category, user experiences differ significantly depending on functional ability,

environment, and coping strategies. Therefore, when utilizing people with disabilities as lead users, issues of representativeness and the need for a multi-layered analytical approach must be carefully considered[100].

While typical persona development projects in practice begin with a sample of around twenty users[73], recruiting participants with disabilities is hindered by factors such as minority status, social and environmental exclusion, stringent ethical requirements, and institutional or caregiver gatekeeping[101]. Also, existing health and social datasets often lack consistent disability indicators or vary in definitions across countries, making comparative measurement difficult[102]. Standardized data collection tools also fail to adequately capture the contextual and subjective aspects of disability experience[103]. Furthermore, in qualitative research involving people with disabilities, communication difficulties, emotional responses, and limited self-expression may further constrain data reliability and interpretability[104]. These limitations—on both quantitative and qualitative fronts—pose structural barriers to achieving representativeness and diversity in disability personas, making them one of the core challenges in this field.

2.4 Service Scenario and Customer Journey map

2.4.1. Scenario

Scenario-based design refers to a set of techniques used in the early stages of development to concretely describe how a service or system is to be used[105]. The scenario in this context is a “narrative” that depicts how a specific user interacts with the service over time to achieve a particular goal within a defined context. This sequence of actions and events unfolds within the usage context, encompassing the user’s goals, plans, and reactions[106].

Scenarios facilitate rapid communication of opportunities and concerns among various stakeholders, which is why designers have long employed them as tools for organizing and communicating ideas[105, 106]. A *service scenario* is typically structured around five key components. First, *setting* refers to the time, place, and environmental background in which the scenario takes place. It defines the physical and social conditions under which the user and the service interact, thereby providing the contextual foundation for service operation and shaping the user experience. Second, *actors* are the users or participants who directly interact with the service. Each actor has distinct goals and is characterized by various attributes such as background, knowledge level, technical proficiency, and personality. In the design process, actors are often embodied as *personas*—fictional yet data-informed user models—to enhance the scenario’s realism. Third, *tools and objects* are the physical or digital artifacts that the actor manipulates or encounters. These include devices, interface elements, data, and

content, all of which serve as enablers or constraints within the service system. Fourth, the *sequence of actions and events* represents the specific steps taken by the actor and the corresponding system or environmental responses over time. This element illustrates how users work toward achieving their goals, and provides valuable insight into potential service bottlenecks or improvement opportunities. Fifth, the *user context* encompasses the underlying reasons and meanings behind the actor's behaviors. It includes internal motivations such as goals, plans, emotions, and reactions, enabling a deeper understanding of users and fostering emotionally attuned design approaches[105].

Scenarios become significantly more effective when they are grounded in personas[105]. Scenarios lacking personas often suffer from low immersion, vague character descriptions, and weak narrative impact. Furthermore, the lack of an empirical foundation reduces their realism and credibility, increasing the risk that scenarios devolve into superficial justifications for specific features or technologies. Such scenarios tend to be rigid and fragmented, which limits their adaptability and scalability[106].

Moreover, a single scenario may lead to vastly different outcomes depending on the user group, underscoring the need to consider multiple personas[107]. To enhance the persuasiveness of a scenario, empirical observation of user behavior is essential. Ethnographic studies, context-based interviews, demographic and market data, and usability testing observations all serve as crucial evidence for grounding scenarios in reality[106].

2.4.2. Customer Journey Map

The Customer Journey Map (CJM) is a conceptual tool that visually represents the process through which a customer engages with a particular service. In its basic form, a CJM illustrates the sequential interactions between the customer and the service provider over time—this sequence is collectively defined as the *journey*. By offering insights into the customer experience, CJMs help organizations develop a deeper understanding of their users, and their practical utility has led to increasing adoption among practitioners[108].

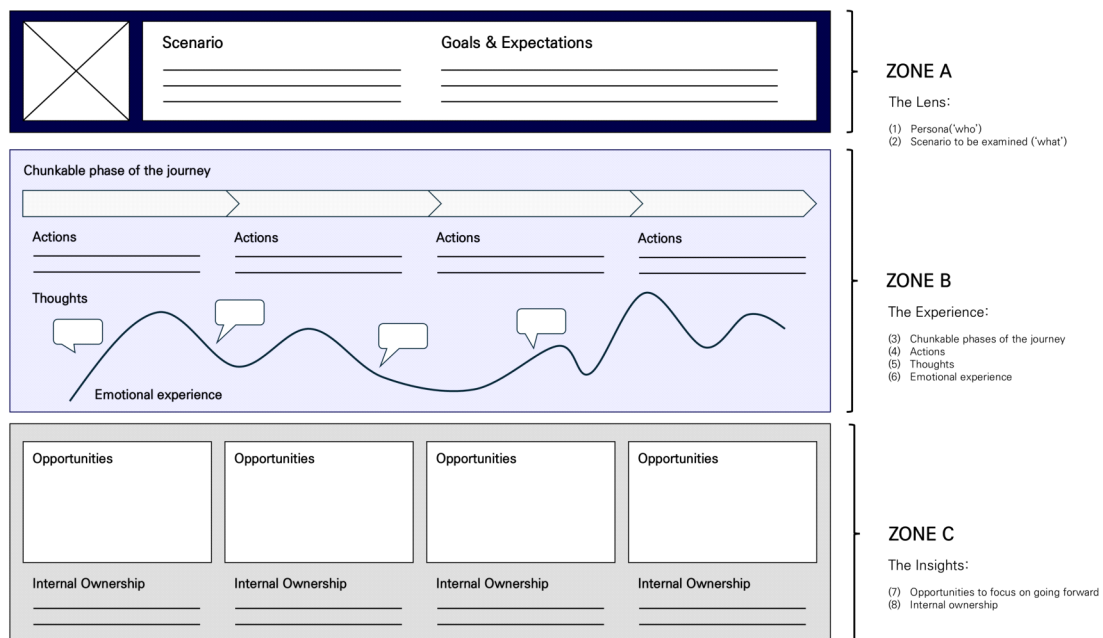


Figure 2-2. Example of Customer Journey Map

Note. Adapted from “Customer Journey Map”, Nielsen Norman Group. Retrieved from <https://www.nngroup.com/articles/customer-journey-mapping/>

As shown in Figure 2-2, CJMs are typically constructed as flow diagrams composed of multiple stages and interlinked touchpoints[109]. Touchpoints refer to moments of interaction that influence customer behavior, which may occur through various mediums such as service personnel, websites, printed materials, devices, other customers, peer groups, or physical environments[110].

The design of the vertical axis in a CJM is a critical factor that determines its effectiveness as an innovation tool. While some models focus solely on visualizing touchpoints, neglecting the importance of the vertical axis, this approach limits the tool's strategic potential. Other approaches recommend configuring the vertical dimension as an emotional journey—capturing intangible customer experiences such as thoughts, beliefs, emotions, and feelings. However, this too may be insufficient for capturing the complexity and interdependence embedded in service systems. A more advanced perspective interprets the vertical axis as a strategic space that integrates managerial components such as design opportunities, customer goals, employee tasks, branding strategies, and omnichannel operations. Yet, as the vertical dimension expands, so does the complexity of the map—leading to potential confusion. Therefore, managers should reconceptualize the vertical axis not as a simple idea repository, but as a structured design layer that illustrates how core organizational functions—marketing, HR, operations, and IT—collaboratively fulfill customer expectations at each touchpoint[111].

Traditionally, CJMs have been developed through qualitative methods such as monitoring, observation, interviews, focus group interviews and surveys, which reconstruct the customer experience[112]. However, with the rise of digital interaction, vast volumes of behavioral log data have become available, enabling

customer behavior to be analyzed and visualized quantitatively[54]. Data-driven CJMs utilize methods such as clustering, process mining, sequence analysis, and deep learning-based behavioral prediction models. These techniques allow for the objective extraction and analysis of real customer journeys, including paths, frequencies, bottlenecks, and drop-off points. Compared to assumption-driven, designer-led interpretations, this data-driven approach offers greater objectivity and reproducibility. As a result, data-driven CJMs are gaining attention as scalable and automatable tools—particularly effective in digital service environments involving large customer bases[113].

2.5 Service Design for People with Developmental Disabilities

2.5.1. Developmental Disabilities

Developmental disabilities are a heterogeneous group of lifelong conditions defined by physical or mental impairments that manifest before the age of 22. They are generally associated with abnormalities in brain or sensory functions and include genetic disorders that affect cognition, behavior, and multiple physiological systems[114, 115]. This category encompasses various conditions such as autism spectrum disorder (ASD), attention-deficit/hyperactivity disorder (ADHD), cerebral palsy, hearing impairments, intellectual disabilities, learning disabilities, visual impairments, and other developmental delays. These conditions typically originate during childhood and result from impairments across physical, cognitive, linguistic, and behavioral domains[116]. The causes and impact of developmental disabilities differs considerably among individuals, even if they share the same diagnosis. For instance, among adults with developmental disabilities, while some adults may require support for daily self-care and independent living, others show relatively strong daily functioning but continue to face difficulties in navigating social interactions and relationships[115]. Although approximately 1.74% of the global population is estimated to have a developmental disability, the broad and continuous nature of the spectrum makes it difficult to precisely characterize individuals, as personal traits differ considerably[24, 117].

Nonetheless, people with developmental disabilities often experience limitations

in fundamental aspects of independent functioning and daily life[29]. The U.S. Administration on Developmental Disabilities (ADD) defines developmental disabilities as conditions that result in substantial limitations in at least three of the following areas of major life activity: (1) self-care (e.g., dressing, bathing, eating), (2) expressive and receptive language, (3) learning, (4) mobility, (5) decision-making, (6) independent living, and (7) economic self-sufficiency[114]. These limitations persist across the lifespan, making people with developmental disabilities less likely to live independently compared to their same-age peers, and typically requiring ongoing support from family and related institutions throughout their development[118]. Paradoxically, despite facing greater risks due to cognitive challenges and limited self-advocacy, people with developmental disabilities are often excluded from policy priorities relative to other disability groups[119]. This exclusion stems from a lack of scientific understanding, as well as a structural bias in existing research and policymaking, which have primarily focused on high-income countries and specific populations[120].

Although institutional and methodological barriers limit the ability to directly assess the service experiences and satisfaction of people with developmental disabilities, existing studies suggest that nondisabled individuals tend to perceive the quality of life of disabled people more negatively than it actually is. This implies that the needs of people with developmental disabilities cannot be fully captured through third-party observation or interpretation alone. Therefore, in order to provide truly necessary and effective services, it is essential to make concerted efforts to directly collect and reflect the perspectives and lived experiences of people with developmental disabilities themselves[121].

2.5.2. ADL/IADL: Functional Indicators for Support Needs in Daily Life

People with developmental disabilities often experience substantial limitations in *adaptive functioning*, including cognition and communication, which lead to dependence on others in various aspects of daily life[118, 121]. These limitations manifest in difficulties with performing daily living skills such as self-care, household activities, and community participation[122]. However, publicly accessible and high-quality instruments for measuring the severity of developmental disabilities and associated daily functioning remain limited[115]. Among these, the most foundational assessment tools for daily functioning are categorized into two domains: Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL)[29]. The concept of *daily living activities* was first introduced by Katz in the 1950s in the fields of geriatrics and health services[116]. ADL includes basic physical and personal hygiene activities such as bathing, toileting, dressing, grooming, and eating. To measure less severe functional limitations, Lawton and Brody developed the complementary IADL index, which includes tasks required for independent living at home and in the community, such as meal preparation and cleanup, housework, laundry, shopping, money management, medication use, telephone use, and transportation[29, 123, 124]. While the number of items included in ADL/IADL measures has varied slightly over time, the core components have remained consistent. These items serve as a classical metric to quantify support needs in Personal Assistance Services (PAS) and Long-Term Support Services (LTSS)[124].

Lawton and Brody also proposed that ADL and IADL activities can be arranged hierarchically based on neuropsychological complexity. The most basic

level consists of essential self-care tasks such as hygiene and mobility (ADL), followed by intermediate activities related to household and community-based independence (physical IADL), and finally the most complex level, which includes social functioning. This hierarchy suggests the possibility of integrating ADL and IADL into a single continuum, but such integration presumes *unidimensionality*—i.e., that both sets of items reflect a common underlying psychological construct. This assumption has been a point of debate in the literature[124].

Some researchers argue that IADLs should be conceptualized as a separate dimension due to their higher neuropsychological complexity and greater susceptibility to cultural and environmental influences[124, 125]. The most common two-dimensional conceptualization divides IADLs into physical IADLs and cognitive IADLs. Physical IADLs include mid-level self-management tasks such as housework and using transportation, while cognitive IADLs encompass higher-order tasks requiring executive functioning, such as financial management, medication use, and telephone communication[125]. Other studies propose that cognitive IADLs should constitute a third dimension, distinct from both ADLs and physical IADLs. Alternatively, some research supports integrating all ADL/IADL items into a single scale or combining physical IADLs with ADLs[124].

LaPlante suggested that measuring assistance needs using ADL/IADL items is applicable across all adult age groups, and that this approach may offer a more comprehensive and sensitive measure of support needs than using an ADL scale alone[124]. Reflecting this dimensional debate, the integration of ADL and IADL in functional disability measurement can be classified into three major

approaches. First, the two domains are treated as separate constructs with independent scales. Second, a single summary score is calculated by incorporating IADL data only when the respondent shows no impairment in ADLs. This enables the identification of moderate disabilities; however, because IADL information is included only when ADL impairments are absent, it systematically excludes cases where ADL impairments are present—presenting a structural limitation in capturing full functional needs. Third, ADL and IADL information are fully integrated to generate a psychometrically valid functional disability score for each respondent[123].

2.5.3. Vineland-3: A Standardized Measure of Adaptive Behavior

Adaptive behavior refers to “the collection of conceptual, social, and practical skills that individuals learn and perform in daily life”. More broadly, it denotes the level of routine task performance required for an individual to fulfill normative roles within society[126, 127]. The presence or absence of deficits in adaptive behavior is considered a core criterion for diagnosing various psychological and psychiatric disorders. For instance, a diagnosis of intellectual disability requires both deficits in general intellectual functioning and impairments in at least one of the three adaptive skill domains—conceptual, social, or practical. In contrast, individuals with autism spectrum disorder commonly show deficits in social and conceptual skills, while practical skills may remain intact[127].

The Vineland Adaptive Behavior Scales, Third Edition (Vineland-3), is a representative standardized assessment tool designed to evaluate multiple

dimensions of adaptive behavior in individuals from birth to over 90 years of age[126]. According to Vineland-3, daily living skills are categorized into three subdomains: the Personal domain for self-care, the Domestic domain for household maintenance, and the Community domain for functioning in school and community settings. These collectively encompass both ADLs and IADLs, and serve as a key metric for assessing functional levels in everyday life[128].

Originally proposed by Doll and later revised by Sparrow and colleagues, the Vineland Scales have evolved into the current three-tiered structure, comprising 11 subdomains aggregated into general and specific adaptive behavior scores across different formats. The initial edition of the Vineland Scales included four adaptive behavior domains: Communication, Socialization, Daily Living Skills, and Motor Skills. The Motor domain is subdivided into Gross and Fine Motor subdomains, while the remaining nine subdomains are distributed across the other three domains[127]. Subsequently, a Maladaptive Behavior domain was added to evaluate problem behaviors; however, this domain is optional and is not used in the calculation of composite scores[129, 130].

The assessment structure of Vineland-3 is illustrated in Figure 2-3[127]. At the highest level, the Adaptive Behavior Composite (ABC) score represents the total score across the entire instrument and quantifies the individual's overall adaptive functioning. This composite is derived from three domain scores: Communication, Daily Living Skills, and Socialization. The Motor Skills domain is excluded from scoring for individuals aged six years and older.

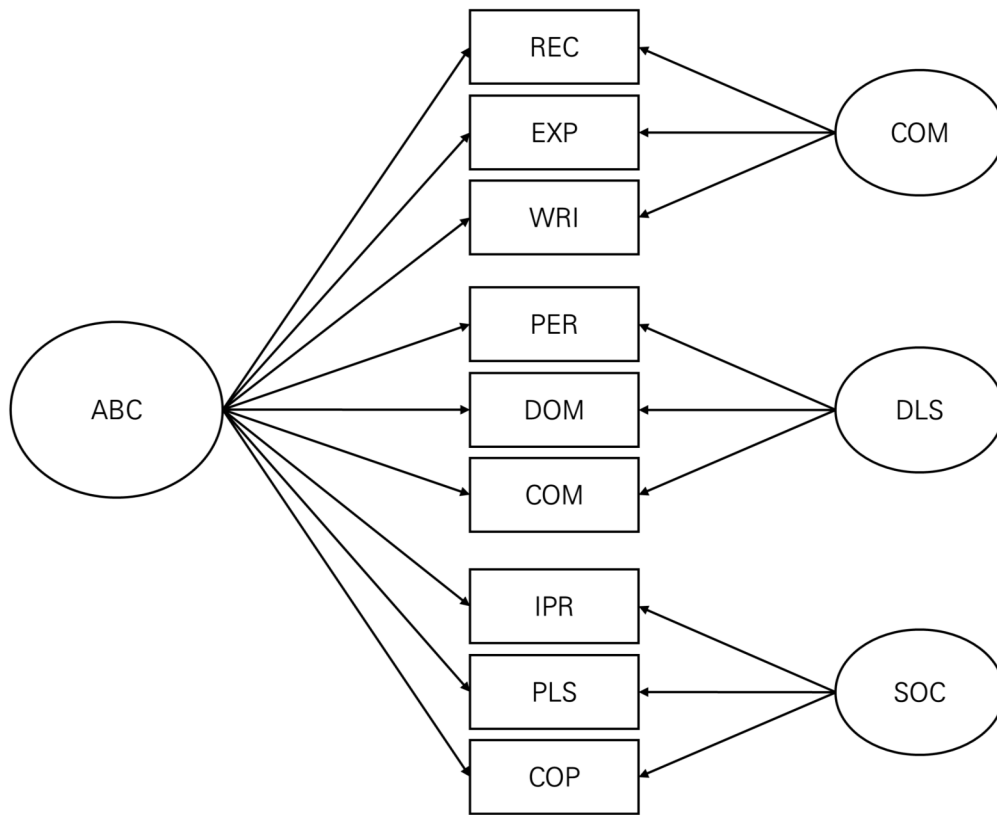


Figure 2-3. Bifactor model of Vineland-3

Note. Adapted from *Is the Vineland-3 Comprehensive Interview form a multidimensional or unidimensional scale? Structural analysis of subdomain scores across early childhood to adulthood*, by Farmer, R. L., Floyd, R. G., & McNicholas, P. J. (2021), *Assessment*.

Table 2-7 Composition of Vineland-3

Domain	Subdomain	Content category	Description
Communication	Receptive	Understanding, listening and attending, following directions, etc.	The ability to listen to and comprehend verbal input, and respond appropriately.
	Expressive	Pre-linguistic behaviors, initiation of speech, conversation, etc.	The ability to verbally express one's thoughts and needs, and appropriately respond to questions.
	Written	Beginning reading, reading skills, writing skills, etc.	The ability to understand letters and compose words and sentences appropriately in writing.
Daily Living Skills	Personal	Eating and drinking, toileting, dressing, hygiene, etc.	Skills related to physical independence and personal care, such as hygiene, eating, and dressing.
	Domestic	Kitchen tasks, house cleaning, etc.	The ability to independently perform household routines such as tidying up and simple chores.
	Community	Telephone use, money management, moving between places, etc.	Practical skills necessary to function in community settings, such as using the phone, managing money, and navigating public spaces.
Socialization	Interpersonal	Responding to others, social communication, etc.	Basic social behavior skills necessary for forming relationships, including responsiveness, empathy, and appropriate communication.
	Play and Leisure	Playing, sharing and cooperating, games, etc.	The ability to engage in play, games, and leisure activities.
	Coping Skills	Manners, apologizing, impulse control, etc.	The ability to self-regulate and respond responsibly to stress or problems in everyday situations.

Each major domain is subdivided into three subdomains, yielding a total of nine subdomain scores. The Communication domain includes the subdomains of Receptive, Expressive, and Written. The Daily Living Skills domain comprises Personal, Domestic, and Community subdomains, which respectively assess competencies related to self-care, household and home management, and money management and community interaction. The Socialization domain includes Interpersonal Relationships, Play and Leisure, and Coping Skills, measuring behavioral traits associated with social interaction and relationship building. Each subdomain is further divided into content categories, which are used to quantify specific aspects of functioning[126, 127].

The Vineland assessment is administered in English and is widely used in various fields across Western countries, including the United States. However, in countries with different cultural contexts, the instrument cannot be used in its original form. Instead, it requires translation and cultural adaptation followed by localization and standardization. Examples include the Korean K-Vineland-II, the Greek VABS II-Gr, and the Vietnamese VVABS. This process involves rigorous validation to ensure the tool's reliability and validity, which can take a considerable amount of time[131-133]. Consequently, in Korea, the Vineland-II remains in use under the localized name K-Vineland-II, while the Korean version of Vineland-3 (K-Vineland-III) is still undergoing standardization. Specifically, K-Vineland-II was published in 2015, while the original Vineland-3 was released in 2016, and the results of the standardization study for K-Vineland-III were only published in 2025. This suggests a time lag of approximately ten years in the development and dissemination of assessment tools for developmental disabilities in Korea compared to other countries.

K-Vineland-II is available in two formats: the Comprehensive Interview Form and the Parent/Caregiver Form. The Comprehensive Interview Form involves a semi-structured interview between a clinician and the caregiver, while the Parent/Caregiver Form is completed directly by the caregiver. Although the administration methods differ, the content of the items remains identical. Each item is scored on a three-point scale, which reflects the level of independence with which the individual performs the specified behavior[130, 131, 134-136].

In addition to ADL/IADL assessments and the Vineland Scales, there are several other instruments available to evaluate functional ability and support needs in people with developmental disabilities, as summarized in Table 2-8[137-140]. However, this study selected the K-Vineland-II as the primary evaluation tool due to its comprehensive life-span coverage, capacity for structured analysis of daily functioning, and ability to quantify adaptive behavior using a standardized scale.

Table 2-8. Standard Tools for Developmental Disabilities

Method	Explanation
Activities of Daily Living Skills (ADL)	A metric that evaluates basic physical functioning such as bathing, grooming, toileting, dressing, and eating, representing the most fundamental level of daily living skills.
Instrumental Activities of Daily Living Skills (IADL)	A supplementary metric that assesses independent living abilities within the home and community, including meal preparation, grocery shopping, housework, laundry, medication management, handling finances, using the telephone, and going out.
Vineland Adaptive Behavior Scales (VABS)	A standardized tool for assessing adaptive behavior in a structured manner. It quantifies daily functioning across the full life span, encompassing domains such as communication, daily living skills, and socialization, and integrates both ADL and IADL components.
Supports Intensity Scale (SIS)	Designed to assess support needs across various areas of daily life. It evaluates the frequency, daily time requirement, and type of support needed across six major life domains: Home Living, Community Living, Life-long Learning, Employment, Health and Safety, and Social Activities.
Autism Diagnostic Observation Schedule (ADOS)	A semi-structured standardized behavioral observation tool used to assess autism spectrum disorders. Examiners evaluate the examinee's core autism-related traits—such as social communication, social responsiveness, and restricted and repetitive behaviors—through structured interactive tasks. The test is composed of five modules selected based on the individual's language ability and age.
Autism Diagnostic Interview-Revised (ADI-R)	A semi-structured interview tool conducted with a primary caregiver to assess children and adults suspected of having autism spectrum disorders or other neurodevelopmental conditions. It evaluates core characteristics of autism in three domains: social interaction, communication and language, and restricted and repetitive behaviors/interests.
Bayley Scales of Infant and Toddler Development (Bayley-III/IV)	A standardized assessment tool used to identify developmental delays in children aged 1 to 42 months, focusing on three core developmental areas: cognition, language, and motor skills.

2.5.4. Service Needs of People with Developmental Disabilities

Limitations in adaptive functioning serve as a significant barrier to independent living for people with developmental disabilities, necessitating continuous support throughout the lifespan. In particular, during the transition from education to adulthood, the roles of families and institutions become increasingly critical[118]. These support needs vary in response to the progression of disability, modifications in the life situations, and variations in the presence or absence of appropriate support provisions[141].

People with developmental disabilities and their families generally require multi-layered support from a wide array of service providers (e.g., physicians, psychologists, speech-language therapists), institutions (e.g., public health clinics, private care organizations), and sectors (e.g., healthcare, education, social services). This complex service structure is one reason people with developmental disabilities experience higher levels of unmet service needs compared to other populations with special needs[142].

Daily life for individuals involves multiple domains, including employment, housing, leisure, healthcare, and community participation[143–145], and people with developmental disabilities tend to rely heavily on instrumental support across these domains. *Instrumental support* refers to tangible and direct assistance with problem-solving or task execution, including the provision of goods or services. It encompasses physical assistance in daily life, health management, service coordination, financial support, and more. Notably, facilitating access to and utilization of formal supports—paid services provided through public or private sectors—is a crucial function of instrumental

support[141].

However, in the case of people with developmental disabilities, natural and informal supports—primarily from family members and close acquaintances—tend to precede formal services. Natural support refers to unpaid assistance provided by individuals within one’s social network, and 95.1% of adults with developmental disabilities receive such support in at least one domain of life. Among them, 55.6% report receiving support across all major domains including employment, housing, leisure, healthcare, and community life, with primary providers being family members such as parents and siblings[144]. Although people with developmental disabilities require support across a range of environments, most continue to reside in family homes, their own homes, or supported housing settings. Some also live in residential facilities, professional nursing homes, intermediate care facilities (ICF-DD), or developmental centers, yet the home and community remain the predominant settings for their daily lives[146].

Accordingly, formal service delivery systems are often designed to be home- and community-centered. Families play dual roles as both natural supporters and providers of instrumental support, acting as connectors between formal and informal support systems. They offer the foundational basis for people with developmental disabilities to pursue self-determined lives, and without the involvement of these natural supporters, access to formal services through public or private means may be significantly hindered. Through the family’s mediating role, formal and informal supports are integrated, enabling people with developmental disabilities to receive more flexible and tailored support that aligns with their specific needs and preferences[141].

2.5.5. Daily Living Support Services

Support services for people with developmental disabilities are provided in various forms, including day programs, supported employment, assistive technology, home modifications, physical therapy, and speech-language therapy[147, 148]. These services share the common function of facilitating stable daily living for people with developmental disabilities and often overlap with the natural support provided by family members. However, natural supports alone are insufficient to address the complex and ongoing needs of this population, necessitating the expansion of systematic support services and the strengthening of effective coordination mechanisms[143].

A representative institutional mechanism fulfilling this role is the provision of *daily living support services*. These services offer practical assistance across daily activities—including domestic chores, community participation, transportation, and personal care—based on assessments of ADL and IADL capabilities. Aimed at people with disabilities, including those with developmental disabilities, these services support community and home-based living and assist families who provide care at home. In this sense, they function to supplement or formalize the support traditionally offered by natural networks, playing a central role in maintaining continuity in everyday life[141, 144, 146, 149, 150].

The terminology and scope of daily living support services differ by country and welfare system. For instance, terms such as Personal Care Services, Supported Living Services, Long-Term Services and Supports (LTSS), and Home and Community-Based Services (HCBS) are commonly used, with service content varying according to national policies and regional contexts[141, 146, 149,

151, 152].

These services were not initially grounded in academic theory but rather emerged in practice in response to policy needs across different countries. Over time, empirical evaluations and policy impact analyses contributed to the academic theorization and structural analysis of such services[153]. Therefore, daily living support services represent both a formal institutional system and a critical field of practice that directly influences the quality of life and level of independence for people with developmental disabilities. In this study, the term *daily living support services* is used as an umbrella concept encompassing these diverse service types.

These services are primarily delivered by trained personnel known as *Direct Support Professionals (DSPs)*. DSPs are responsible for a wide range of tasks, from assisting with daily activities to promoting physical and emotional well-being. Their duties include managing health and safety, facilitating interpersonal relationships and social networks, supporting communication, assisting with personal hygiene and mobility, fostering self-advocacy and financial management, enabling community engagement, preventing crises, and conducting domestic tasks and self-care training. Despite playing a pivotal role in daily functioning for people with developmental disabilities, DSPs face structural challenges such as low wages, poor working conditions, limited opportunities for professional development, excessive workloads, emotional stress, and low societal recognition[144, 150].

Given that people with developmental disabilities experience diverse levels of limitation in basic functional capabilities required for independent living, the specific types of support needed vary significantly by individual and

developmental stage. For instance, daily living tasks for a three-year-old child may include putting away toys, feeding oneself, or toileting with caregiver assistance, whereas a sixteen-year-old adolescent may be expected to perform simple meal preparation, laundry, and room cleaning. However, not all individuals follow this typical developmental trajectory, and some adults with developmental disabilities over the age of sixteen may still struggle with basic daily living skills[29, 116, 122]. These variations in functional limitation—both in nature and severity—are critical factors for service design, intervention planning, research, and public health policymaking beyond the diagnostic label itself[115].

Therefore, the timely deployment of DSPs according to the individual's functional level is essential. In practice, however, DSPs are often assigned an excessively broad scope of duties, and there is a lack of precise matching between each professional's strengths and the client's specific needs[153, 154]. This mismatch can lead to overburdening, reduced specialization, and ultimately a decline in service quality. To address these issues, a system is needed to quantitatively assess support needs and service domains in daily life, thereby enabling the strategic assignment of DSPs with domain-specific expertise. Such an approach would improve efficiency for both service recipients and providers, enhance DSPs' professional capacities, and reduce their workload.

Chapter 3. Methodology Description

This study proposes a process that augments a limited set of data from individuals with developmental disabilities to generate personas and quantitatively identify needs for daily living support services by disability type. Current disability service design remains largely provider-centric, lacking frameworks that systematically reflect users' actual needs and contextual behaviors. For people with developmental disabilities, standardized surveys are often unsuitable due to cognitive and communicative limitations, and the spectrum nature of these disabilities leads to significant variability even among those with the same diagnosis. Consequently, both the quantity and quality of available data are limited, and small interview samples must often represent highly diverse needs—challenging the applicability of persona-based approaches. To address this challenge, a quantitative methodology is needed—one that can augment a small set of data to extract representative behavior patterns and service needs. Based on this recognition, this study adopts a data-driven service design approach and sets forth the following two research questions:

RQ #1. Can representative personas be generated through meaningful clustering of limited interview data from people with developmental disabilities?

RQ #2. Can the specific needs for daily living support services of people with developmental disabilities be quantitatively assessed?

3.1 Augmented Persona Mapping Framework

To address the research questions, this study designed a three-step process: (1) data augmentation based on actual interview data from individuals with developmental disabilities, (2) clustering based on behavioral characteristics and derivation of representative personas, and (3) mapping of daily living support service needs by persona type. The overall research process workflow is illustrated in Figure 3-1. Through this, the study proposes an *Augmented Persona Mapping Framework* that enables consistent and reproducible identification of “service needs” among people with disabilities.

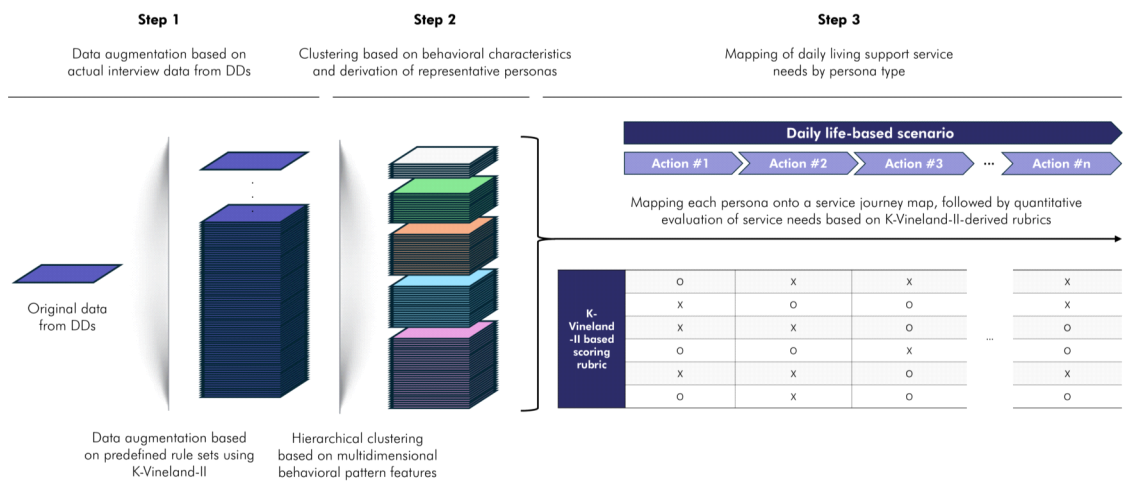


Figure 3-1. Overall Research Workflow

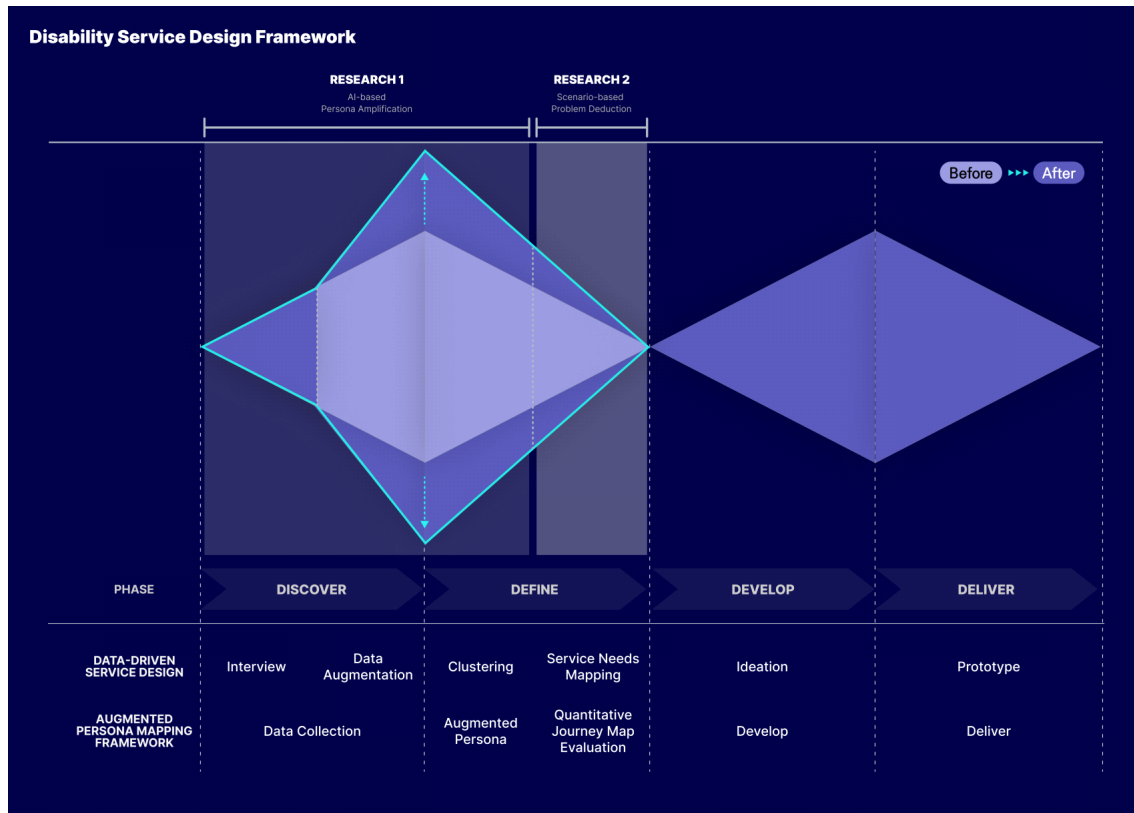


Figure 3-2. Augmented Persona Mapping Framework

Figure 3-2 visually illustrates the framework's process of data augmentation, convergence into representative personas, and validation of service needs within the context of disability personas. Typically, personas are not only descriptive of user characteristics but also intertwined with user behaviors and experiences in specific scenarios. However, Salminen et al. (2020) emphasized the importance of structuring personas in consideration of the researcher's and/or practitioner's goals, as well as the layout and categorization of persona information[68]. Reflecting this perspective, the overall structure of this study adopts the Double

Diamond model of service design. Specifically, the first diamond's "Discover" and "Define" phases are reinterpreted as Part 1 and Part 2 of the framework, corresponding to persona generation (Part 1) and service need analysis (Part 2). This segmentation aims to clarify the logical distinction and connection between these two core components of the study.

3.1.1. Part 1: Data Augmentation and Persona Clustering

Part 1 describes the process of generating representative personas through data augmentation. In practice, collecting detailed qualitative data on the behavioral characteristics of people with developmental disabilities is highly constrained. The number of respondents tends to be limited, and because reliable data require behavior-based observation, the research process often entails significant time and financial costs. To overcome these limitations, this study utilized interview data based on the K-Vineland-II to segment adaptive behavior abilities and augment the dataset by applying a predefined rule set that reflects the characteristics of developmental disabilities. The rule set used in this study was structured based on the scoring and classification logic of the K-Vineland-II, a standardized assessment tool for adaptive behavior in people with developmental disabilities.

The rationale for using the K-Vineland-II lies in its ability to quantitatively and structurally assess the multidimensional and spectrum-based nature of developmental disabilities. This tool comprehensively evaluates adaptive behavior and allows behavioral performance to be structured and scored in relative terms, making it particularly suitable for data augmentation and pattern analysis. While

the Vineland-3 is the most recent standardized version internationally, the K-Vineland-II was selected because it is the most up-to-date version standardized for use in Korea, where the study was conducted. Nevertheless, because the K-Vineland-II relies on proxy reporting by caregivers rather than direct observation, subjectivity may influence the ratings, potentially affecting data reliability. The tool also reflects task performance within specific contexts (time, space, and environment), limiting its generalizability to all real-world service scenarios. Furthermore, its three-point rating scale may introduce ambiguity and boundary issues in interpretation.

Despite these limitations, the K-Vineland-II remains the most suitable tool for structurally evaluating adaptive behavior and obtaining quantitative data from a small sample of individuals with developmental disabilities. This study proceeds with the recognition that acquiring large-scale disability data is inherently difficult, and adopts a algorithm-based data augmentation approach to make the most of limited data, while acknowledging that the generated dataset may not fully represent the entire developmental disability spectrum. Accordingly, the composition of personas and the resulting analysis may change if the sample pool is expanded in future studies.

Based on the augmented dataset, clustering analysis was conducted according to disability characteristics. Clustering criteria reflected domain-specific scoring logic and behavioral similarity as defined by the K-Vineland-II. As a result, a set of persona types was derived, each representing a cluster of individuals with similar functional and behavioral profiles. During persona construction, each cluster was aligned with the axes of a predefined disability persona skeleton to ensure conceptual consistency across the framework.

3.1.2. Part 2: Quantitative Mapping and Evaluation of Service Needs

Part 2 involves quantitatively assessing the service needs of each representative persona, as defined in Part 1, across various daily living scenarios. For people with developmental disabilities, service utilization is closely linked to challenges in performing everyday tasks, and support needs tend to become particularly evident in specific contexts or activities. Therefore, analyzing service demands by key daily scenarios provides a critical foundation for identifying real-life problems and designing targeted intervention strategies. In this study, the personas are constructed not only to reflect user characteristics but also to embed potential difficulties within specific scenarios, thereby facilitating an empathetic understanding of accessibility issues and enabling practical service planning for people with disabilities[107].

Accordingly, this stage first defines daily scenarios and related service components and then develops a scoring rubric based on the K-Vineland-II to evaluate the required functional capacities. The comparison between each persona's adaptive functioning and the demands of each scenario allows for a quantitative identification of capability gaps and support needs.

3.2 Part 1: Data Augmentation and Persona Clustering

3.2.1. Persona Skeleton for Developmental Disabilities

Persona generation typically proceeds in two stages. First, user data is segmented by analysts or algorithms. Then, the segmented data is reinforced using a foundational structure known as a *persona skeleton*, which is further enriched with personalized information to form a fully developed persona profile. This *personalization* stage involves transforming anonymized, impersonal data segments into identifiable user representations with specific behavioral and demographic traits[91].

Accordingly, a clear definition of the persona skeleton is essential prior to generating personas for people with disabilities. The persona skeleton serves as a structural framework formed in the early stages of persona development, built upon initial assumptions and fragmented findings from preliminary research. It functions as the basis through which the persona's key characteristics and attributes gradually take shape[98]. However, previous studies indicate that even among general personas, there is no standardized definition of the persona skeleton, and definitions concerning disability-specific personas are even more limited. Some prior research, for instance, has addressed the accessibility challenges faced by individuals with various types of disabilities in operating home appliances. These studies proposed disability persona skeletons visualizing accessibility issues based on elements such as demographic information, task barriers and goals, frustrations and needs, representative quotes, and character

representations[59]. Reflecting on critiques that such approaches often amount to little more than adding impairment data to existing persona structures, this study aims to design a new form of disability persona that incorporates everyday context and environmental factors into its foundation.

Demographic Info

Name

Gender

Age (Actual)

Age (Functional)

Lifestyle

Living Situation

Work & School Life

Social

Social Media Usage Patterns

Hobbies & Preferences

Health

Relationship

Family

• Key People

• Releationship

Friends & Others

• Key People

• Releationship

A day in the life

Daily Activities

Community

Leisure

Disability Characteristics

Diagnosis / Presence of Multiple Disabilities

Disability Severity

Disability-related Dispositions

Maldaptive behavior

Disability-related Physical Traits

Coping Capacity

Medications

ABC Level

Communication

Receptive

Score : 75/100 (V Score)

Expressive

Score : 75/100 (V Score)

Writing

Score : 75/100 (V Score)

Daily Living Skills

Personal

Score : 75/100 (V Score)

Domestic

Score : 75/100 (V Score)

Community

Score : 75/100 (V Score)

Socialization

Interpersonal Relationships

Score : 75/100 (V Score)

Play & Leisure

Score : 75/100 (V Score)

Coping skills

Score : 75/100 (V Score)

Comprehensive Review

Figure 3-3. Persona Skeleton for Developmental Disabilities

Figure 3-3 visualizes the structure of the developmental disability persona skeleton proposed in this study, based on the engaging persona model. While the basic layout refers to persona components summarized in Table 2-3, it has been restructured to align with the characteristics of people with developmental disabilities and the objectives of this research.

The proposed persona skeleton consists of three main categories. First, the *Demographic Info* section includes not only basic demographic details such as name, age, gender, and personality, but also contextual information on daily life – such as “Lifestyle,” “Relationship,” and “A Day in the Life.” These additions aim to reflect the individual’s everyday routines and environments, providing narrative depth that goes beyond a list of isolated traits to reveal behavioral context and environmental influences. Second, the *Disability Characteristics* section encompasses diagnostic information, severity and specific traits of the disability (e.g., sensory sensitivities, communication styles), coping skills, and medication usage. This category provides a comprehensive account of disability-related features and personal specificities gathered through interviews. Third, the ABC Level section presents a quantified profile of functional levels across nine subdomains of daily living skills, based on the K-Vineland-II framework. Each domain is scored on a 100-point absolute scale, complemented by V-scale scores indicating relative performance. Additionally, natural language interpretations of these scores – generated using large language models – are presented alongside, enabling a mixed-format profile that combines quantitative metrics with qualitative narratives. This dual approach allows for an intuitive understanding of each persona’s functional profile and service needs.

This persona skeleton framework moves beyond simple profiling to offer a

standardized structure that supports user-centered service scenario design and accessibility assessment.

3.2.2. Interview Design for Data Collection

To construct a foundational dataset for generating personas of people with developmental disabilities, this study designed and conducted semi-structured interviews. The primary objective was to obtain in-depth information on each participant's demographic profile, personality traits, coping skills, lifestyle, daily context, and degree of disability—ultimately enabling the development of digitally replicable personas.

The interview was organized into two parts. Part 1 aimed to comprehensively capture the life context of people with developmental disabilities and was adapted from the interview script used by Park et al. (2024) in constructing personas that reproduced real individuals' tendencies and decisions with approximately 85% similarity. This script was derived and simplified from the American Voices Project, which collects everyday life data representative of the U.S. population[95]. The American Voices Project employs open-ended questions across the following domains: (1) life history and major events, (2) rhythms of daily life, (3) family and social relationships, (4) experiences and responses to financial hardship, (5) income and expenditure patterns, (6) household members' participation in formal and informal labor, (7) public welfare utilization, (8) housing and community environment, (9) health and healthcare usage, (10) experiences related to mental health, substance use, and stress, (11) political orientation and civic engagement, and (12) core identity and sources of

meaning[155]. From these domains, this study selected and tailored items most relevant to the characteristics of people with developmental disabilities for inclusion in Part 1.

Part 2 was designed to obtain a more detailed and quantitative understanding of disability severity, using the comprehensive interview form of the K-Vineland-II assessment. This tool enables the objective evaluation of adaptive functioning in areas such as communication, daily living skills, and socialization. The interview format was chosen over the parent/caregiver format to minimize the influence of subjective judgments by informants. In addition to the standard 3-point scale (unable to perform, partially able, fully able) used in K-Vineland-II, the interviews incorporated follow-up questions to capture actual performance levels, degree of difficulty, and specific situational challenges for each item. This approach aimed to obtain more reliable accounts of observable behaviors and contextual explanations.

3.2.3. Data Augmentation Algorithm

One of the most significant challenges in constructing personas for people with developmental disabilities is the scarcity of sufficient data. To address this limitation, this study developed a systematic data augmentation methodology based on algorithmic principles. Rather than relying on random generation, the approach applies carefully designed constraints that reflect the functional characteristics and developmental patterns of developmental disabilities.

The augmentation process was conducted using responses to the 307 items from the K-Vineland-II. To maintain the essential characteristics of

developmental disabilities while ensuring adequate variability, two core constraints were introduced. First, the NOISE_RATIO parameter limited each generated variation to within a 10% deviation from the total of the original response. Second, to accurately reflect the developmental dependencies between K-Vineland-II items, a set of dependency rules was constructed (see Figure 3-4). Although the K-Vineland-II items are arranged in order of difficulty based on performance within a standardized population, items within the same content category are not always sequential or grouped[130]. Accordingly, this study restructured the *dependency rules* by modeling the inherent inter-item relationships and hierarchical structure—reflecting the developmental principle that higher-order functions presuppose the mastery of foundational skills. This ensured that the augmented data would realistically simulate behavioral patterns observed in people with developmental disabilities. Additionally, based on the characteristics of the K-Vineland-II, the algorithm identified specific items for which partial performance (a score of 1) was logically infeasible and encoded these as rules within the algorithm.

For each original dataset, 400 variations were generated, resulting in an expanded dataset 400 times the size of the original. This provided a robust foundation for exploring diverse developmental disability profiles from a limited set of real-world cases.

K-Vineland-II: Receptive Language, Expressive Language, Written Language, Personal Living, Domestic Living, Community Living, Interpersonal Relationships, Play and Leisure, and Coping Skills. These scores were then converted into standardized V-scale scores (ranging from 1 to 24), which allow for comparison of an individual's relative functional level against age-matched peers—a key standard scoring system used in the K-Vineland-II[130]. An algorithmic mapping was implemented to convert raw domain scores into V-scale scores.

Based on the average V-scale scores, the data were first classified into three hierarchical levels: High ($V\text{-scale} \geq 12.0$), Moderate ($5.0 \leq V\text{-scale} < 12.0$), and Low ($V\text{-scale} < 5.0$). A second-stage clustering was then conducted within each hierarchical level to identify more granular behavioral patterns. This analysis went beyond simple score comparison and incorporated statistical guidance from the K-Vineland-II manual to extract multidimensional behavioral pattern features, as summarized in Table 3-1.

Table 3-1. Extracted Multidimensional Behavioral Pattern Features

Multidimensional Behavioral Pattern Features	Detailed Description
Response Pattern Characteristics	Distribution (proportions of 0, 1, 2), variability, transition patterns, persistence
Domain Profile Characteristics	Relative strengths/weaknesses, profile shape, imbalance indicators
Functional Area Patterns	Interrelations among communication, daily living skills, and socialization domains

These multidimensional features were reduced using PCA and subsequently clustered using a hierarchical clustering algorithm based on Ward's linkage

method. The optimal number of clusters was determined by synthesizing silhouette scores and target file distribution. Each cluster was assigned a unique three-digit ID, combining its hierarchical tier and cluster number.

The clustering results were validated through t-SNE visualizations and radar charts. Additionally, Z-score analysis by domain was conducted to quantify the relative strengths and weaknesses of each cluster. This allowed for a systematic typology of diverse functional profiles within the population of people with developmental disabilities.

3.2.5. Final Construction of Representative Personas

In the final stage, representative personas were constructed by integrating the characteristics and behavioral pattern analyses of each cluster into the persona skeleton framework defined in Section 3.2.1. The construction process for each cluster's persona skeleton followed the steps below:

(1) Integrated Quantitative–Qualitative Analysis by Domain

The V-scale scores across the nine major domains were normalized into percentages, and then transformed into descriptive narratives using a large language model. This process incorporated both the numerical scores and their relative strength levels (e.g., “very strong,” “weak”) to analyze the behavioral characteristics in each domain.

(2) Creation of a Comprehensive Functional Profile

Based on the aggregated scores and features of each domain, a holistic

narrative of the overall functional profile was generated. The narrative encompassed patterns of strength and weakness across domains, imbalances or harmonies among major functional areas, and their implications for daily life and support strategies.

(3) Integration into Persona Template

The quantitative scores and qualitative narratives were synthesized into the persona skeleton template. Since no established template previously existed for personas specific to developmental disabilities, the template was extended to incorporate disability-specific characteristics, such as the severity of autism and coping capacity.

The resulting persona skeletons offer a comprehensive and multidimensional understanding of the functional characteristics of people with developmental disabilities by organically combining clinical assessment results with real-life contexts. Furthermore, the use of large language models enabled data-driven, objective analysis while enhancing the contextual richness of the descriptive narratives.

3.3 Part 2: Quantitative Mapping and Evaluation of Service Needs

3.3.1. Daily Living Scenario Design

To validate the ecological validity of the developmental disability types identified through clustering, this study assessed the functional performance of each cluster within daily living contexts. The aim of this evaluation was to objectively examine how theoretical functional profiles manifest in real-life environments.

To this end, a set of daily living scenarios was constructed to encompass representative activity domains that people with developmental disabilities commonly encounter. To develop the horizontal axis of these scenarios, prior studies that reflect the actual daily lives of people with developmental disabilities were reviewed[111]. However, while there were scenario-based assessments focused on specific domains—such as school life or cooking—few studies offered a structured definition or a comprehensive list of daily activities. Some studies selected only partial domains arbitrarily for the purpose of behavioral intervention, and others, despite covering a broader scope, did not disclose sufficient detail about their scenario composition, limiting the potential for analysis[118, 122, 128, 156].

In response, this study constructed a multidimensional daily living scenario framework that reflects ADL/IADL, grounded in both relevant prior research and empirical data obtained through interviews with individuals with developmental

disabilities and their caregivers. The scenario consists of six core domains—Self-Care, Meal, Mobility, Community, Activity, and Household Tasks—each encompassing concrete tasks and their contextual variations, as detailed in Table 3-2.

Table 3-2. Representative Daily Living Scenarios

Domain	Scenarios
Self-Care	Toileting, toothbrushing, face washing, showering, dressing
Meal	Cooking, eating, taking medication
Mobility	Using buses and subways
Community	Various situations at school
Activity	Shopping, hospital visit
Household Tasks	Cleaning, doing the dishes, doing laundry

Each domain was further broken down into specific behavioral units, with explicitly defined functional requirements for completing each task. This structure provides a robust foundation for objectively analyzing variations in task performance among people with developmental disabilities across diverse real-life contexts.

3.3.2. Quantitative Scoring and Visualization of Journey Map

While existing assessment tools such as ADL/IADL and Vineland-3 have been widely used to evaluate the daily living skills of people with developmental disabilities, few approaches have quantitatively measured performance feasibility

in specific real-life contexts through automated methods. To address this limitation, the present study introduced a quantitative scoring system into the scenario-based journey map defined earlier and designed a visualization process accordingly.

First, the horizontal axis of the journey map was constructed based on the previously defined *daily living activities* and their corresponding performance elements. The vertical axis was composed of all items from the K-Vineland-II. Each functional element required for task performance was mapped to specific items, resulting in a scoring matrix for evaluating functional ability. This process went beyond simple item-to-activity matching by applying *dependency rules* that account for prerequisite skills and developmental hierarchies. Accordingly, each task was associated not only with the key skills it directly requires, but also with foundational abilities that precede them developmentally. Because the evaluation was based on each representative persona's K-Vineland-II responses derived from Part 1, this design ensured both validity and individualization of the scoring.

Subsequently, task performance scores were calculated for each cluster. For every task, the mode value of the related items was used to determine a standardized score on a scale of 0 (unable), 50 (partially able), and 100 (fully able), and the mean score was computed accordingly. The results were visualized using color coding—red (<50), yellow (50 - 89), and green (≥ 90)—to enable intuitive interpretation. In particular, tasks with low scores were further analyzed using a large language model (GPT-4o). This analysis was strictly limited to the provided item content and modal responses in order to maintain objectivity.

Through this scoring process, the study was able to identify the strengths and weaknesses of each cluster in specific daily contexts, offering insights beyond mere functional scores to actual behavioral characteristics.

Chapter 4. Validation of Augmented Persona Mapping Framework

4.1 Overview of Ground Truth Data Collection

4.1.1. Interview-Based Dataset Preparation

This study conducted a quantitative assessment of the functional characteristics and daily behavioral patterns of people with developmental disabilities to build a foundational dataset for persona construction. To this end, semi-structured interviews were conducted with eight individuals diagnosed with developmental disabilities. Participants were selected to reflect diversity in age, gender, disability type, and severity. In accordance with the K-Vineland-II guidelines, interview respondents were adult caregivers (e.g., parents or siblings) who shared daily life with the individuals and were well-positioned to provide detailed observations on their behaviors and functional abilities[130]. All respondents were informed in advance that the collected data could be used for the development of AI models simulating human behavior and assessing the functional performance of people with developmental disabilities in specific daily scenarios; informed consent was obtained accordingly. The interviews followed the two-part structure designed in Section 3.2.2—covering contextual information and K-Vineland-II-based functional assessments—and each session lasted an

average of 93 minutes. As a result, a comprehensive dataset was constructed for each individual, including demographic information, lifestyle patterns, relationship characteristics, and responses to all 307 functional items. These data were organized according to the persona skeleton framework defined in Section 3.2.1, resulting in eight initial persona profiles.

Table 4-1. Interview Target Profiles

Participant	Gender	Chronological Age	Adaptive Age	Diagnoses	Interview Duration
A	F	24	3	Intellectual Disability	98 min
B	F	22	9	Intellectual Disability	128 min
C	M	23	9	Intellectual Disability	73 min
D	M	22	15	Intellectual Disability, Autism, ADHD	97 min
E	M	22	10	Intellectual Disability	74 min
F	M	22	11	Autism	91 min
G	F	23	10	Intellectual Disability, Autism, Cerebral Palsy, Visual Impairment	105 min
H	F	22	13	Intellectual Disability, Autism, Cerebral Palsy, Visual Impairment	76 min

The eight collected personas were initially evaluated through profile analysis based on the K-Vineland-II. Scores were calculated for each of the nine key domains—Receptive Language, Expressive Language, Written Language, Personal Living, Domestic Living, Community, Interpersonal Relationships, Play & Leisure, and Coping Skills—and were standardized for comparative analysis. The functional features of each profile were interpreted multidimensionally, based on inter-domain performance gaps, relative strengths and weaknesses, and

item-level dependency patterns. Cosine similarity analysis was used to quantitatively assess the similarity between profiles, revealing that the eight cases could be grouped into three performance tiers: high, moderate, and low. As shown in Figure 4-1, while the overall cases were divided into three functional levels, domain-specific strengths and weaknesses varied within each group, revealing notable differences in patterns even among profiles of the same tier. This finding confirmed that simple classification by overall performance level is insufficient to fully capture the diversity and complexity of developmental disabilities functional profiles. Therefore, this preliminary analysis highlighted the need for multidimensional clustering rather than relying on a unidimensional scoring system. Additionally, one of the eight cases displayed a distinct profile that significantly diverged from the others in both domain scores and pattern structure. This case was identified as a potential outlier and may serve as a test case for evaluating extrapolation validity in subsequent validation phases.

Vineland Domain Profiles (A-H)

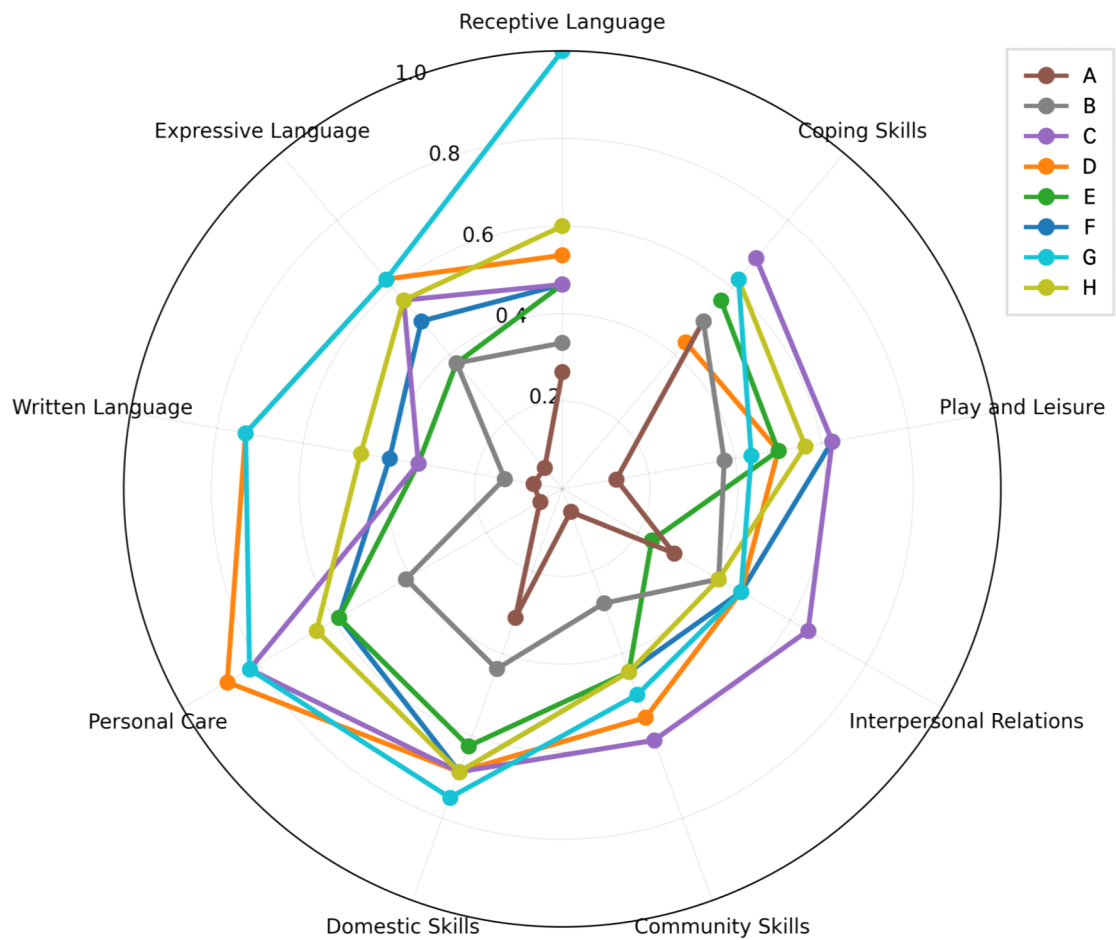


Figure 4-1. Functional Domain Profiles of Interview Targets

4.2 Part 1: Construction and Evaluation of Representative Personas

4.2.1. Augmenting Base Profiles for Persona Modeling

To ensure statistical representativeness of the functional diversity and behavioral characteristics of people with developmental disabilities, this study applied an algorithm-based persona augmentation methodology. As outlined in Section 3.2.3, this approach aims to generate a large volume of augmented data by applying controlled variations while preserving the structural integrity of original cases.

Among the eight base profiles, six were selected for augmentation, excluding one case that exhibited pronounced profile idiosyncrasies and one case that showed the most typical pattern in the mid-functionality group. For each selected case, 400 variations were generated, resulting in a total of 2,406 augmented persona datasets, including 6 original datasets.

Shin et al. (2024) have noted that large language models tend to generate unrealistic or inconsistent characteristics when creating personas based on user data, primarily due to their inability to accurately replicate statistical distributions[76]. To address this issue, this study excluded LLM-based generation and instead adopted a structured, numerically grounded augmentation method to ensure both reliability and controllability. The augmentation process employed a constraint-based algorithm that strictly adheres to the dependency rules among items in the K-Vineland-II. Specifically, under the assumption that

the ability to perform higher-order functional tasks requires competence in prerequisite foundational skills, the algorithm generated response patterns that respect developmental hierarchies. Analysis of the statistical distributions and pattern similarities in the generated dataset confirmed that the core characteristics of developmental disabilities were generally preserved.

4.2.2. Hierarchical Behavioral Clustering of Augmented Profiles

For the 2,406 augmented persona datasets, this study conducted a multi-layered clustering procedure as described in Section 3.2.4. First, V-scale scores were calculated for each of the nine domains for every persona, and the average score was used to determine the overall functional level. Based on the average V-scale scores, the data were categorized into three tiers: High ($V\text{-scale} \geq 12.0$), Medium ($5.0 \leq V\text{-scale} < 12.0$), and Low ($V\text{-scale} < 5.0$). Within each tier, multidimensional behavioral pattern features—such as response distributions, domain profile shapes, and relative strengths and weaknesses across functional areas—were extracted and vectorized, serving as input for hierarchical clustering. Euclidean distance and Ward’s linkage method were used as the distance metric and clustering algorithm, respectively.

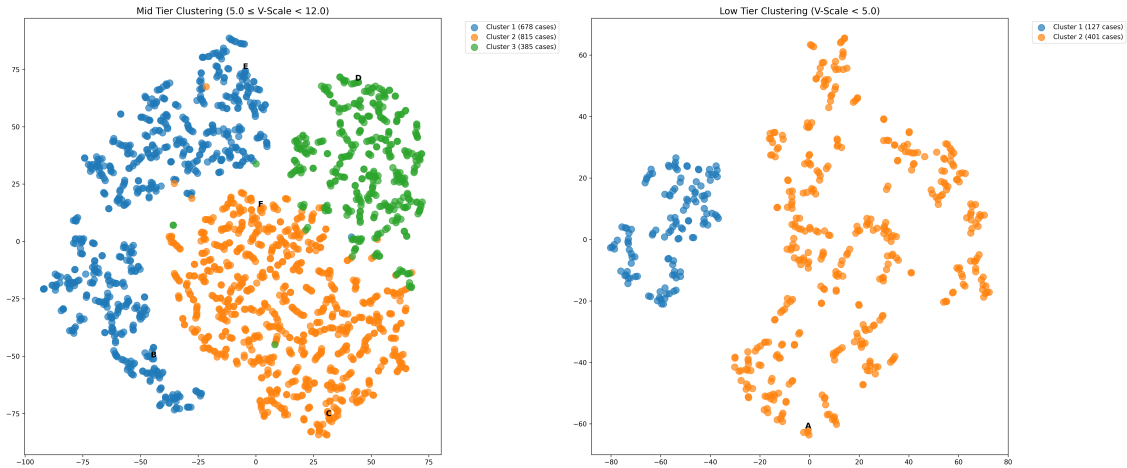


Figure 4-2. Clustering Result

As a result, five meaningful developmental disability profile clusters were identified (three from the medium tier and two from the low tier). The validity of the clustering results was evaluated using silhouette scores and t-SNE visualizations. Each cluster achieved an average silhouette coefficient of 0.62, indicating satisfactory cohesion and separation. Furthermore, t-SNE-based dimensionality reduction visualization showed clear boundaries between clusters. A one-way analysis of variance (ANOVA) on domain scores across clusters yielded statistically significant differences, $F(4,2401) = 27.21$, $p < .001$. To ensure comparability across domains, all domain scores were converted to a standardized 100-point scale. This allowed for intuitive visualization and comparison of relative functional performance trends across clusters.

4.2.3. Functional Pattern Characterization of Persona Clusters

Each of the five clusters derived from the clustering results exhibited the following functional characteristics and performance patterns:

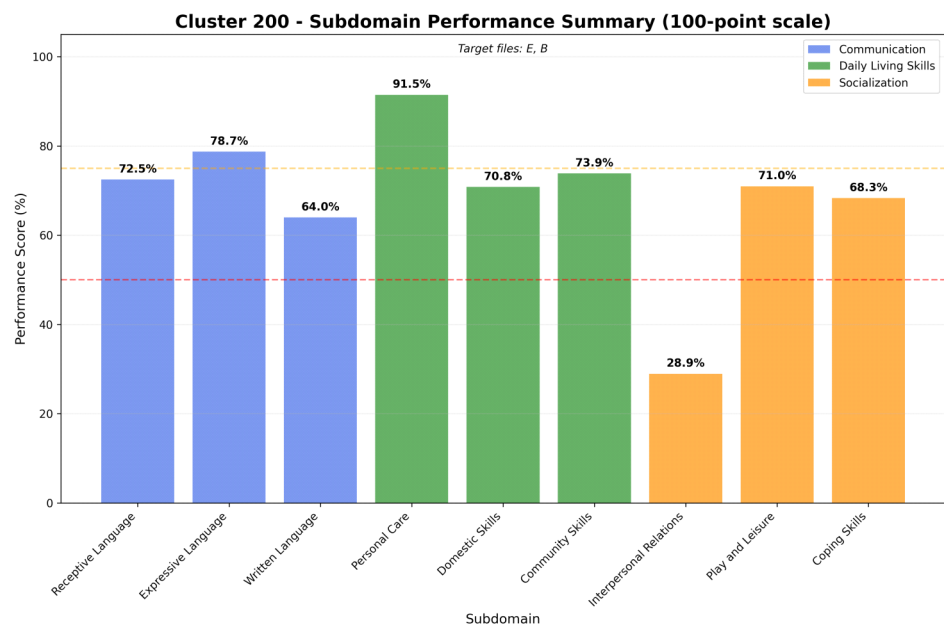


Figure 4-3. Subdomain-level Functional Profile of Cluster 200

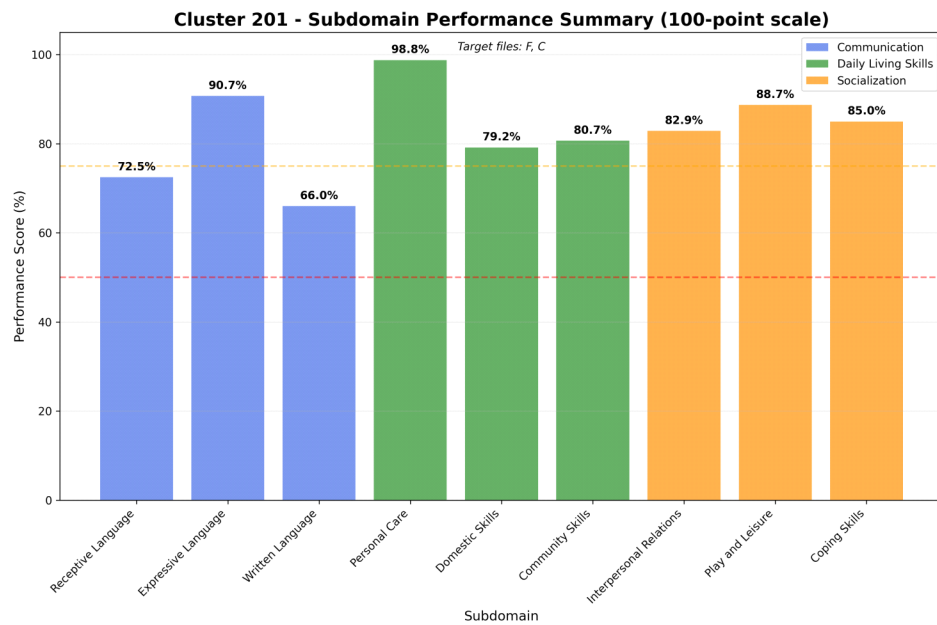


Figure 4-4. Subdomain-level Functional Profile of Cluster 201

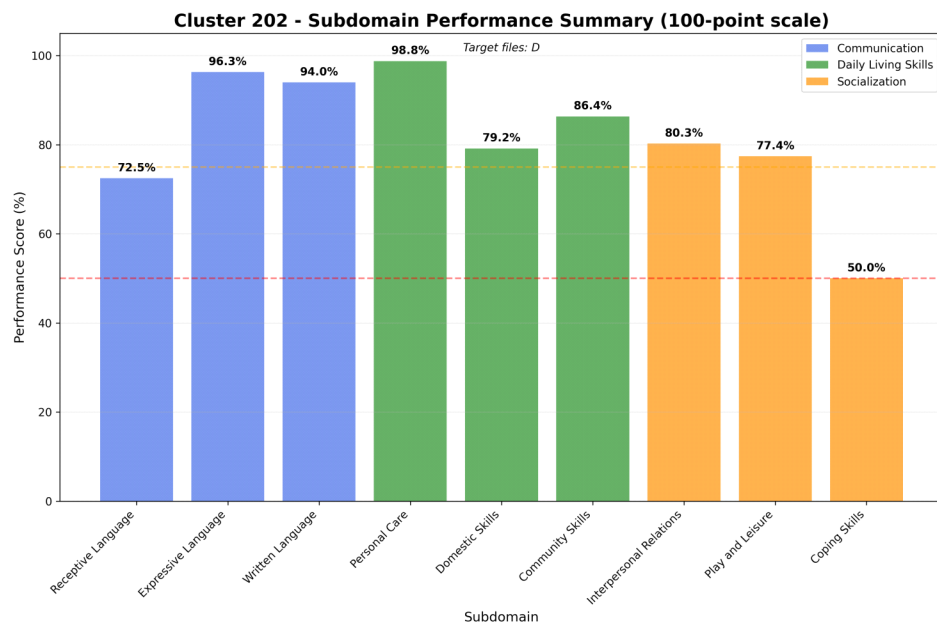


Figure 4-5. Subdomain-level Functional Profile of Cluster 202

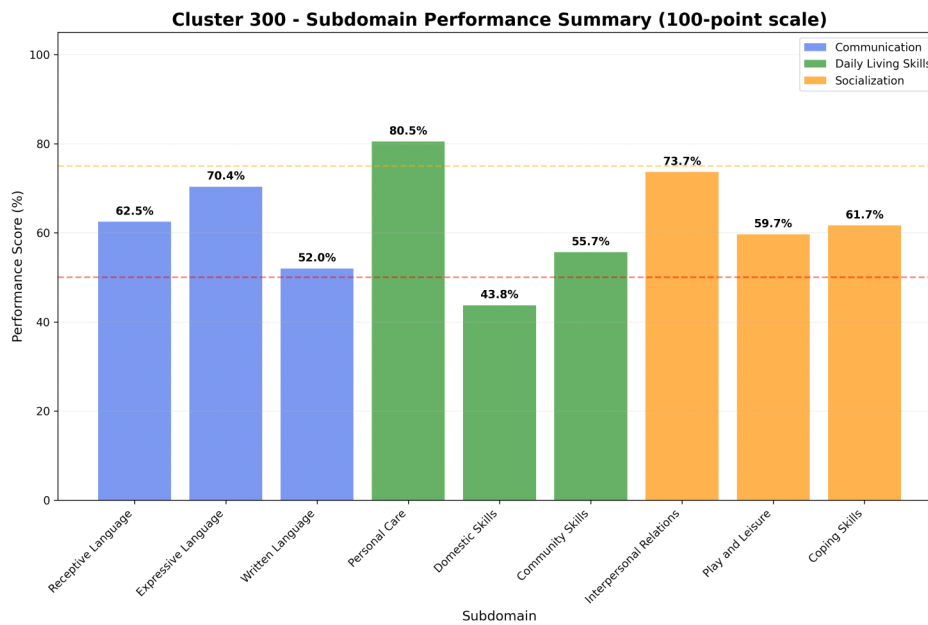


Figure 4-6. Subdomain-level Functional Profile of Cluster 300

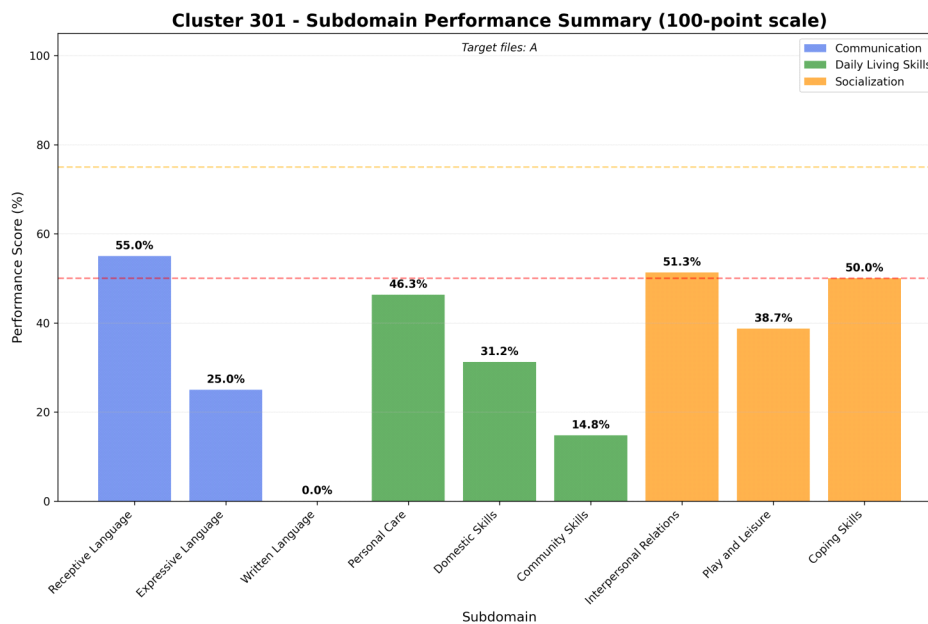


Figure 4-7. Subdomain-level Functional Profile of Cluster 301

- Cluster 200

Cluster 200 demonstrated high overall performance, with particularly strong abilities in the domain of personal living. In contrast, scores in interpersonal relationships were relatively lower, suggesting a profile inclined toward individualism, with strengths in self-management and independent functioning rather than social interaction. Original profiles B and E fall into this cluster.

- Cluster 201

Cluster 201 also showed consistently high performance across all domains, with especially strong skills in expressive language and personal living. Although this cluster also performed well in general socialization domains, written language appeared relatively weak. This group corresponds to a well-balanced high-functioning profile, with original profiles C and F classified here.

- Cluster 202

Cluster 202 exhibited exceptional performance in expressive language, written language, and personal living domains but showed relative weaknesses in domestic living, receptive language, and play and leisure. This indicates internal variability in functional capacity. Notably, the coping skills domain scored as low as 50%, suggesting low adaptability to changing situations or complex tasks. Original profile D corresponds to this cluster.

- Cluster 300

Cluster 300 showed about half the performance level compared to the top three clusters. While it demonstrated strong self-management skills in personal living, it exhibited significant difficulties in domestic living and community participation, reflecting a functional imbalance. Communication and socialization skills were moderate overall, with interpersonal relationships being a relative strength. This cluster is interpreted as having established independence in routine daily tasks, but limited adaptability to social contexts or environmental changes. Notably, this cluster type did not appear in the original dataset and was newly identified through data augmentation.

- Cluster 301

Cluster 301 represents the most extreme group, showing the lowest performance levels overall. In particular, expressive language (25.0%), written language (0%), and community participation (14.6%) scored extremely low. However, receptive language and socialization domains were relatively higher, suggesting that while some capacity for receptive information processing and social engagement remains, the ability to express and execute is severely limited. Original profile A corresponds to this cluster.

To move beyond a simple relative comparison of developmental levels and precisely identify what functional abilities each cluster entails, it is necessary to analyze the results at the level of K-Vineland-II's content categories (totaling 48 areas). Although these items can be visualized as in Figure 4-8, the complexity and interpretive difficulty of the information present limitations for practical use.

Therefore, in the next phase of persona generation, this study seeks to utilize large language models to more efficiently interpret and integrate these detailed functional items.

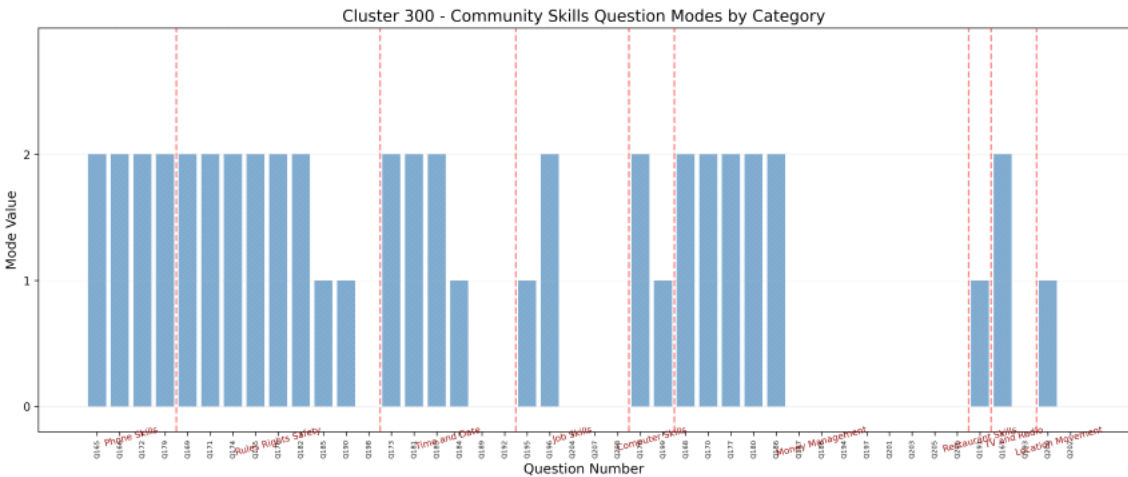


Figure 4-8. Example of Functional Analysis by Content Category

4.2.4. Validation of Persona Cluster Fidelity

To validate the credibility of the five developmental disability profiles derived through clustering and contextual analysis, and to assess their applicability to real-world cases, a validation experiment was conducted using two cases that had been excluded from the initial analysis. These two cases had been pre-classified as representing an atypical pattern (G) and a typical pattern (H), respectively. Based on their K-Vineland-II response data, each case’s alignment with the existing clusters was evaluated.

The primary criterion for determining alignment was the response match rate

between the full set of K-Vineland-II item responses for each validation case and the mode-based response data of each cluster. To ensure a more precise evaluation of similarity, the following three composite indicators were applied:

- (1) Weighted Similarity: This metric accounts for the three-point scale structure of the K-Vineland-II by considering 0 and 1, as well as 1 and 2, to represent adjacent levels of functioning, while 0 and 2 are treated as distinctly different performance levels. This approach provides a more nuanced reflection of subtle differences in functional capacity.
- (2) Cosine Similarity: This measures the overall shape similarity between the response vectors of the validation case and the representative cluster profile. It emphasizes relative pattern similarity rather than absolute values, thus evaluating how closely the overall structure of profiles aligns.
- (3) MAE Similarity (Mean Absolute Error Similarity): This indicator calculates the average difference in item scores to assess the actual numerical proximity. A lower mean error indicates a higher degree of similarity in response values.

As a result, case G showed the highest similarity with Cluster 202. The detailed evaluation of G's alignment showed a response match rate of 75.6%, a cosine similarity of 92.4%, and a weighted similarity of 78.7%, aligning with the characteristics of the mid-functioning cluster exhibiting high internal functional variability. On the other hand, case H showed 77.5% similarity with Cluster 201.

As a profile pre-identified as having a typical pattern, H demonstrated a high degree of alignment with Cluster 201, which included the largest number of datasets. These findings confirm that the clustering model developed in this study can be reliably applied to real-world cases.

Additionally, the domain-level alignment between each cluster's domain structure and the actual responses from validation cases showed that the model could also predict domain-level differences. Notably, case H exhibited consistent responses across all domains, while case G showed a deviation pattern in the coping skills and socialization domains that closely resembled the average profile of Cluster 202. These validation results support both the robustness and generalizability of the clustering methodology proposed in this study.

Table 4-2. Verification of Cluster Affinity for Case 'G'

Similarity Analysis with All Clusters				
Cluster	Match Rate	Weighted Similarity	Cosine Similarity	MAE Similarity
Cluster 200	61.9%	65.9%	85.3%	71.8%
Cluster 201	73.3%	77.3%	92.9%	83.2%
Cluster 202	75.6%	78.7%	92.4%	83.4%
Cluster 300	58.3%	68.3%	85.5%	70.8%
Cluster 301	33.6%	37.6%	62.9%	43.6%
Detailed Similarity Analysis with Cluster 202				
Domain	Subdomain	Matched Items	Match Rate	Cosine Similarity
Communication	REC	13 / 20	65.0%	87.4%
	EXP	49 / 54	90.7%	98.0%
	WRI	22 / 25	88.0%	96.7%
Daily Living Skills	PER	38 / 41	92.7%	99.1%
	DOM	16 / 24	66.7%	91.7%
	COM	34 / 44	77.3%	90.4%
Socialization	IPR	30 / 38	78.9%	95.2%
	PLS	19 / 31	61.3%	84.6%
	COP	11 / 30	36.7%	74.6%

Table 4-3. Verification of Cluster Affinity for Case 'H'

Similarity Analysis with All Clusters				
Cluster	Match Rate	Weighted Similarity	Cosine Similarity	MAE Similarity
Cluster 200	67.4%	71.3%	88.0%	77.2%
Cluster 201	77.5%	80.9%	93.9%	86.0%
Cluster 202	73.0%	76.8%	92.1%	82.6%
Cluster 300	62.9%	67.7%	87.2%	74.9%
Cluster 301	37.1%	41.4%	63.8%	47.7%
Detailed Similarity Analysis with Cluster 202				
Domain	Subdomain	Matched Items	Match Rate	Cosine Similarity
Communication	REC	14 / 20	70.0%	95.4%
	EXP	49 / 54	90.7%	98.7%
	WRI	20 / 25	80.0%	94.4%
Daily Living Skills	PER	35 / 41	85.4%	98.1%
	DOM	17 / 24	70.8%	95.2%
	COM	36 / 44	81.8%	91.5%
Socialization	IPR	26 / 38	68.4%	84.4%
	PLS	25 / 31	80.6%	94.1%
	COP	16 / 30	53.3%	90.6%

Thus, the five clusters derived through clustering analysis were validated as representative personas that can accurately reflect the characteristics of people with developmental disabilities. However, as noted in Section 4.2.3, simply presenting item responses poses interpretative challenges. In particular, the K-Vineland-II presents items not sequentially within a single content category but in a discontinuous structure, mixing various content areas depending on the developmental level. This structural limitation impairs consistency in result interpretation.

To address these interpretative difficulties, this study utilized the ABC Level component of the previously defined developmental disability persona skeleton to visualize the developmental levels of each representative persona. Functional

performance characteristics within each domain were organized according to the content categories of the K-Vineland-II. Furthermore, a large language model was employed to generate summaries and interpretations that reflected semantic connections between items and contextual nuances of functional performance. To ensure that the LLM-based analysis did not produce distorted or biased results, the researcher conducted a manual cross-validation of all summaries by reviewing the original data. This process confirmed that there was no semantic distortion or loss of information. As a result, each representative persona’s core characteristics were visualized in an intuitive format, enhancing both interpretability and practical usability.

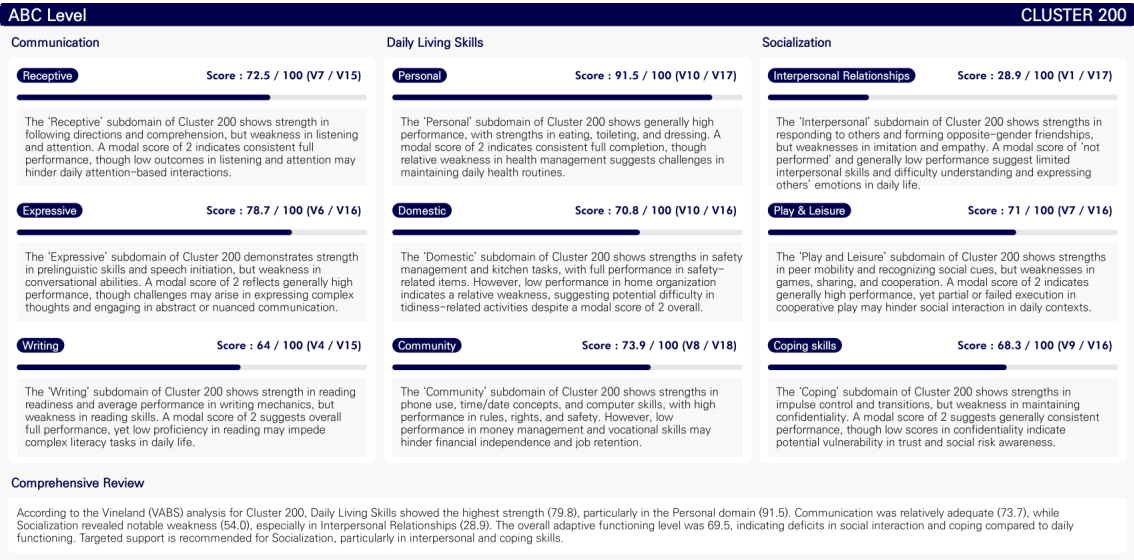


Figure 4-9. Representative Persona Report: Cluster 200

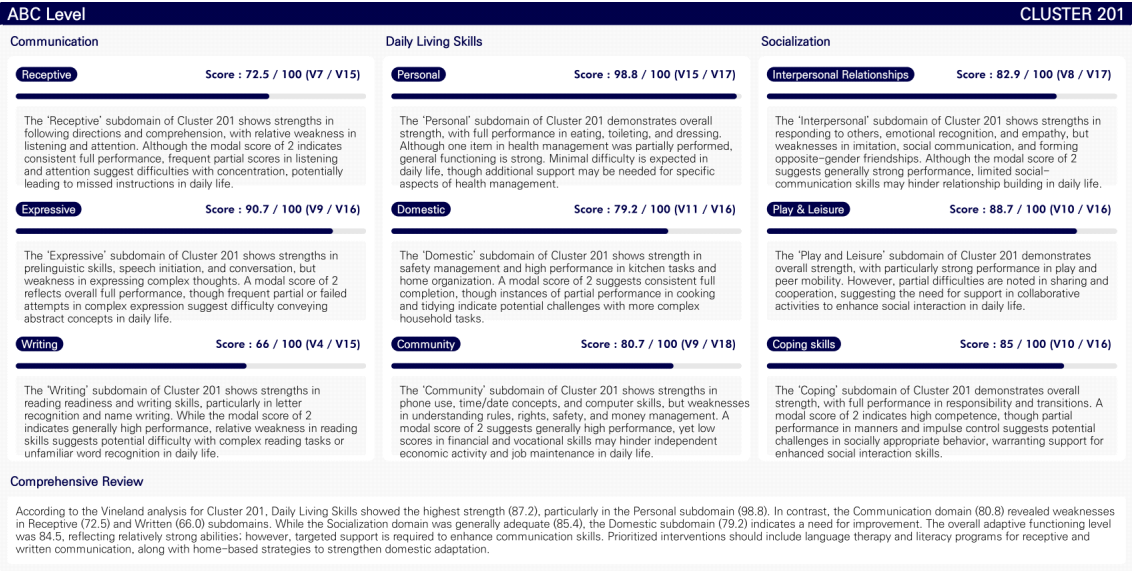


Figure 4–10. Representative Persona Report: Cluster 201

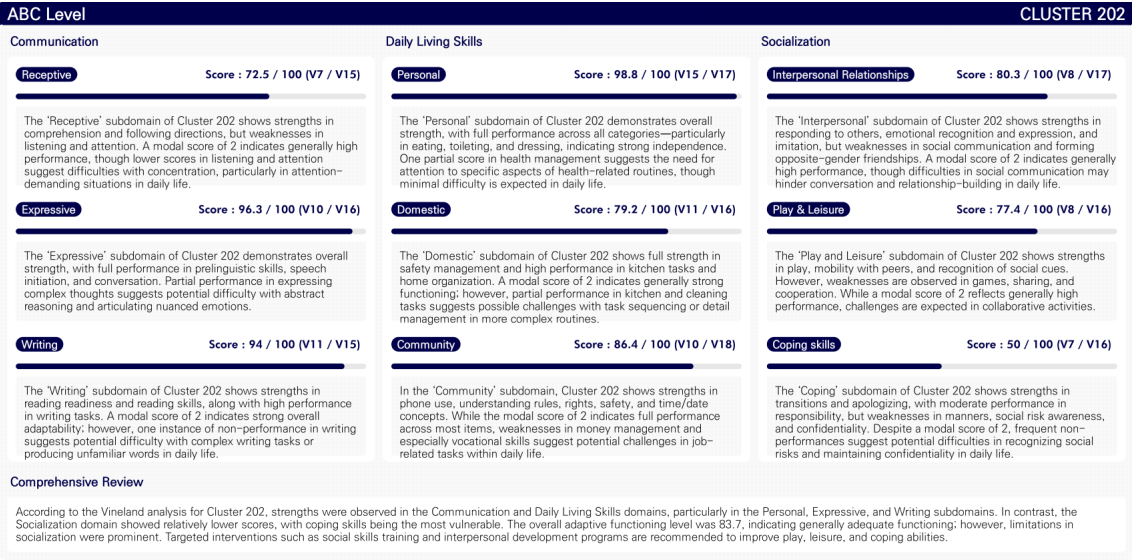


Figure 4–11. Representative Persona Report: Cluster 202

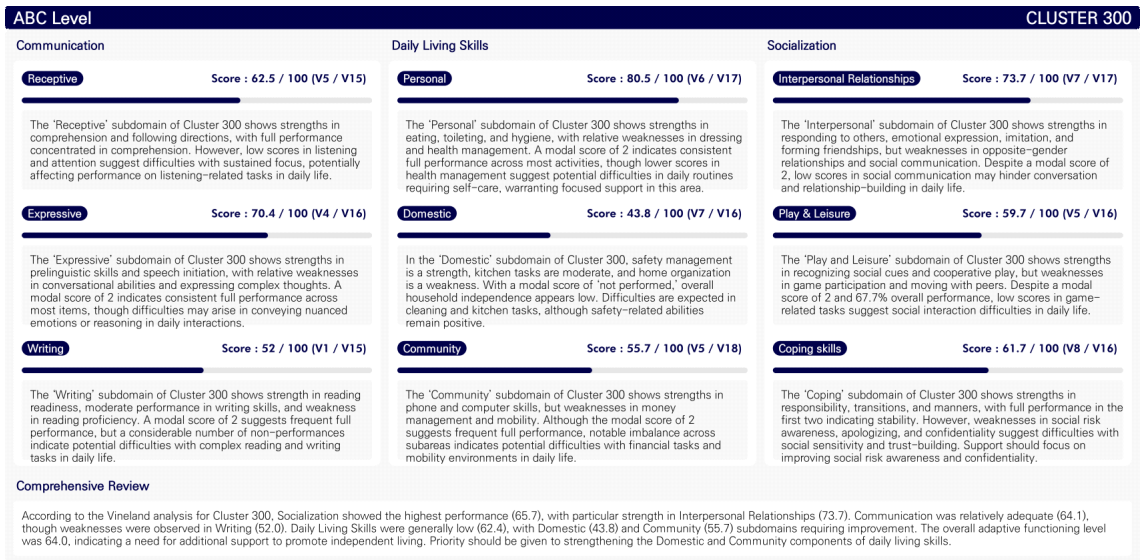


Figure 4-12. Representative Persona Report: Cluster 300

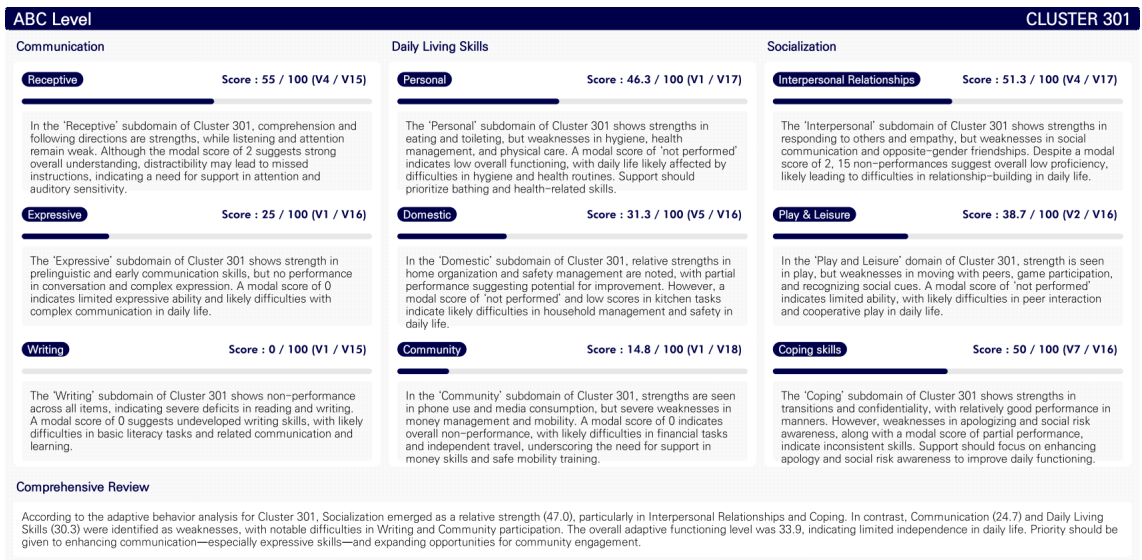


Figure 4-13. Representative Persona Report: Cluster 301

The constructed representative personas were evaluated to accurately reflect behavioral patterns in real-life daily contexts while encompassing the diverse functional profiles of people with developmental disabilities. This suggests their potential as a valuable tool for gaining a multidimensional understanding of the complex characteristics of people with developmental disabilities—beyond simple classification by functional level—and for providing more tailored support.

4.3 Part 2: Evaluation of Daily Living Support Needs Using Persona-based Journey Maps

4.3.1. Quantitative Mapping of Daily Living Scenarios

Based on the clustering results, this study applied the daily living domain-based assessment framework introduced in Section 3.3.1 to concretely analyze and compare the performance capabilities of representative personas in everyday contexts. According to the Vineland-3 framework, daily living activities occur within three major contexts—Personal, Domestic, and Community. Building on this, six subdomains were defined for this study: Self-care, Meal, Mobility, Community, Activity, and Household Tasks.

While conventional service journey maps tend to focus on specific service touchpoints or usage scenarios—often limiting the analysis to fragmented views of the service itself—in designing services for people with developmental disabilities, it is essential to first assess challenges and functional capabilities within the broader continuity of daily living. Only through such comprehensive analysis can practical pain points be identified and areas requiring intervention be structurally understood.

Accordingly, this study subdivided daily living into 11 concrete scenarios and defined a total of 135 detailed activity items for evaluation. These detailed items were derived from prior research. For example, studies have shown that *grocery shopping* involves a series of sub-activities such as making a shopping list, locating items, comparing prices, communicating with staff, and making

payments[157].

The performance outcomes were organized to provide a multi-dimensional view of the functional patterns exhibited by each representative persona. Using the previously developed journey map-based automated scoring system, scenario-specific performance levels were calculated. These quantified outcomes were structured to visually reveal each cluster's relative strengths and vulnerabilities. Rather than emphasizing precise score comparisons, the analysis focused on identifying behavioral patterns, dominant tendencies, and prioritizing domains requiring support.

Table 4-4. Scenario-based Performance Evaluation (Personal Domain)

Self Care - Toileting					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Move to the restroom	100	100	100	100	100
Prepare the toilet	100	100	100	100	100
Unfasten pants button	100	100	100	66.7	16.7
Unzip pants	100	100	100	100	100
Lower pants and underwear	100	100	100	100	100
Sit on the toilet and void	100	100	100	100	100
Wipe with toilet paper	100	100	100	100	100
Flush the toilet	100	100	100	100	100
Pull up pants and underwear	100	100	100	100	100
Zip up pants	100	100	100	100	100
Fasten pants button	100	100	100	83.3	33.3
Move to the sink and turn on the water	66.7	83.3	100	100	33.3
Wash hands	50	75	100	100	50
Apply soap to hands	50	75	100	100	50
Rinse hands	50	75	100	100	50
Dry hands with a towel	100	100	100	100	50
Self Care - Toothbrushing					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Pick up a toothbrush and toothpaste	100	100	100	100	0
Apply toothpaste to the toothbrush	100	100	100	50	0
Brush teeth with the toothbrush	100	100	100	100	0
Brush the tongue with the toothbrush	100	100	100	100	0
Rinse the mouth	100	100	100	100	16.7
Rinse the toothbrush	100	100	100	100	0
Wipe the mouth area with a towel	100	100	100	100	50
Self Care - Face washing					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Dispense facial cleanser onto the hands	100	100	100	50	25
Wash the face	100	100	100	100	25
Rinse the face	100	100	100	100	50
Dry the face with a towel	100	100	100	100	50

Self Care - Showering					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Turn on the faucet and adjust water to an appropriate temperature	100	100	100	100	25
Wet the hair and body	100	100	100	100	16.7
Dispense shampoo	100	100	100	100	16.7
Apply shampoo to the hair	100	100	100	100	16.7
Rinse the hair thoroughly	100	100	100	100	16.7
Dispense facial cleanser onto the hands	100	100	100	100	16.7
Wash the face	100	100	100	100	16.7
Rinse the face	100	100	100	100	16.7
Dispense body wash	100	100	100	100	16.7
Apply body wash to the body	100	100	100	100	16.7
Rinse the body thoroughly	100	100	100	100	16.7
Turn off the water	100	100	100	100	16.7
Dry the hair and body with a towel	100	100	100	100	16.7
Dry the hair using a hair dryer	100	100	100	100	12.5
Self Care - Dressing					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Check the weather and daily schedule	93.8	100	100	75	12.5
Select appropriate clothing based on the weather and context	100	100	100	0	0
Put on underwear	100	100	100	100	100
Put on pants	100	100	100	100	100
Zip up pants	100	100	100	100	100
Fasten pants button	100	100	100	50	0
Put on a T-shirt	100	100	100	100	100
Put on a shirt	100	100	100	100	100
Button up the shirt	83.3	100	100	50	0
Put on socks	100	100	100	100	100
Put on a jacket	100	100	100	100	50
Zip up the jacket	100	100	100	83.3	50
Put on shoes	100	100	100	100	50
Tie shoelaces	100	100	100	100	50

Table 4-5. Scenario-based Performance Evaluation (Domestic Domain)

Meal					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Review the recipe	88.9	77.8	77.8	55.6	0
Take out ingredients according to the recipe	66.7	33.3	33.3	0	0
Wash the ingredients	100	100	100	100	50
Chop and prepare the ingredients	83.3	83.3	83.3	66.7	0
Heat the soup	80	90	80	50	10
Stir-fry rice	83.3	83.3	75	41.7	8.3
Take out side dishes	100	100	100	100	0
Use the microwave to heat side dishes	75	100	100	75	0
Set the table	100	100	100	100	100
Eat the meal	100	100	100	100	91.7
Take medication	100	100	90	80	60
Clear the table	100	91.7	100	83.3	58.3
Household Tasks					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Organize personal belongings in the room	100	100	100	83.3	83.3
Vacuum the personal room	75	100	100	37.5	62.5
Vacuum the living room	62.5	87.5	75	37.5	75
Pre-rinse dishes with water	90	100	100	80	50
Apply dish soap to the dishes	90	100	100	60	50
Rinse off dishes with clean water	91.7	100	100	66.7	41.7
Add detergent and fabric softener to the washing machine	0	100	100	0	50
Load laundry into the washing machine and start the cycle	66.7	100	91.7	41.7	16.7
Transfer washed clothes to the dryer and start the cycle	66.7	100	91.7	41.7	16.7
Fold the clothes	100	100	100	66.7	83.3
Organize the folded clothes	100	100	100	66.7	83.3

Table 4-6. Scenario-based Performance Evaluation (Community Domain)

Mobility					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Check transportation information to the destination	91.7	95.8	100	66.7	8.3
Walk to the bus stop	100	100	100	75	50
Cross the crosswalk	100	100	100	83.3	16.7
Wait at the bus stop	100	100	100	100	100
Identify the correct bus to the destination	100	100	100	83.3	33.3
Tap the transit card and board the bus	83.3	100	100	100	33.3
Recognize arrival at the transfer stop	100	95.5	95.5	90.9	45.5
Tap the transit card and get off the bus	83.3	100	100	100	33.3
Recognize that one got off at the wrong stop	90	90	60	50	60
Call a guardian to ask for assistance	85.7	89.3	89.3	67.9	21.4
Use a smartphone to find an alternative route	83.3	91.7	100	50	16.7
Identify an alternative arriving bus based on the new route	100	100	100	83.3	33.3
Tap the transit card and board the bus	83.3	100	100	100	33.3
Recognize arrival at the destination	100	95.5	95.5	90.9	45.5
Tap the transit card and get off the bus	83.3	100	100	100	33.3
School Participation					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Greet peers	31.8	100	100	90.9	72.7
Transition to the next class according to the schedule	100	100	66.7	100	100
Sit down at one's seat at the beginning of class	100	100	50	83.3	66.7
Respond when called during roll call	100	91.7	91.7	100	100
Pay attention during class	57.1	50	35.7	42.9	35.7
Read aloud or silently from the textbook	66.7	72.2	100	55.6	0

Take notes on the class material	57.1	53.6	67.9	35.7	10.7
Answer a question when called on by the teacher	75	90	87.5	72.5	40
Discuss group activity topics with classmates	44.7	86.8	73.7	71.1	65.8
Share personal belongings (e.g., pencil, eraser, toy) with peers	28.6	100	78.6	92.9	71.4
Borrow an item from a peer	50	100	50	100	50
Talk with friends during break time	51.8	91.1	80.4	78.6	62.5
Express gratitude when receiving a gift from a peer	58.3	100	100	100	75
Complete a task or errand assigned by the teacher	94.4	88.9	88.9	72.2	72.2
Ask a peer for help	75	100	93.8	100	87.5
Participate in basic individual physical activities (e.g., kicking a ball, vaulting)	83.3	100	100	50	0
Participate in team sports (e.g., soccer, volleyball, basketball)	54.5	68.2	59.1	31.8	0
Record homework or announcements during end-of-day homeroom	95.8	83.3	91.7	70.8	50
Say goodbye to friends and leave school	61.1	100	88.9	100	83.3
Activity - Shopping					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Create a shopping list	100	83.3	91.7	58.3	8.3
Consider the required budget	85.7	92.9	92.9	57.1	7.1
Search for a suitable store	75	100	100	50	50
Travel to the shopping mall	87.5	100	100	75	0
Ask a store clerk for the location of an item	90.5	97.6	76.2	76.2	47.6
Find the needed item from the display shelf	100	100	100	100	12.5

Read price tags and compare items before making a selection	78.6	85.7	92.9	57.1	0
Pay for the item through the clerk at the counter	58.3	66.7	45.8	37.5	33.3
Carry purchased items and return home	87.5	100	100	75	0
Activity - Hospital Visit					
	Cluster 200	Cluster 201	Cluster 202	Cluster 300	Cluster 301
Recognize symptoms such as cold, fever, chest pain, or injury	50	100	100	50	0
Take available medication and recognize if medication is insufficient	66.7	100	83.3	50	0
Search for a nearby clinic or hospital	80	90	100	60	20
Inform a guardian and request assistance	91.2	100	97.1	82.4	29.4
Call to make an appointment at a hospital or clinic	75	87.5	100	62.5	25
Travel to the medical facility	87.5	100	100	75	0
Describe symptoms at the reception desk	70	93.3	96.7	83.3	20
Explain symptoms during a consultation with the doctor	73.5	94.1	97.1	85.3	17.6
Receive a prescription and complete payment	60	75	45	45	40
Go to the pharmacy	81.2	93.8	100	68.8	12.5
Present the prescription to the pharmacist	82.1	92.9	85.7	82.1	17.9
Receive the medication and complete payment	62.5	75	87.5	56.2	0
Return home	87.5	100	100	75	0
Take the prescribed medication	100	100	90	80	60

The analysis revealed that each cluster exhibited distinct patterns of strengths and weaknesses in daily living contexts. For example, Cluster 200 demonstrated strong performance in personal care (average score of 92.4), but showed relative limitations in the Activity (58.1) and Community (53.2) domains. Cluster 201 displayed balanced performance across all areas, with an average score of 75.6 across all scenarios. Cluster 202 performed well in visually oriented tasks (e.g., shopping, face washing), but recorded lower scores in community settings or coping situations that required abstract problem-solving. Cluster 300 was capable of performing routine tasks such as self-care and household tasks, but showed limited adaptability to external environments, with an average community score of only 23.4. Cluster 301 demonstrated very low overall functional capacity; it was virtually unable to perform expression-based tasks and showed only minimal receptive responses.

4.3.2. Qualitative Diagnosis of Support Needs Using LLMs

This study goes beyond the mere visualization of performance scores by redesigning the vertical axis of the journey map to serve as a diagnostic domain that explains the contextual causes of each persona's functional limitations. This approach aims to provide a more nuanced interpretation of the underlying reasons for performance constraints, which are not easily discernible from score differences alone. Furthermore, the activities within each daily living scenario—corresponding to the horizontal axis of the journey map—were aligned with the subdomains of the K-Vineland-II. This restructuring addresses the limitations of conventional journey maps, which are often restricted to fragmented service

touchpoints, and expands the framework to analyze the entirety of daily life through units of adaptive behavior based on the characteristics and capabilities of people with developmental disabilities.

To this end, an in-depth qualitative analysis was conducted using GPT-4o, a large language model, for the performance outcomes across the 11 previously defined daily living scenarios. This analysis followed the evaluation rubric designed in Section 3.3.2, which was based on the K-Vineland-II. The analysis focused on items that fell below performance standards within the rubric, taking into account the interaction among response patterns, inter-domain relationships, and developmental traits specific to each cluster.

LLM responses were generated based on standardized prompts and a consistent response format. To ensure the reliability of the analysis, the researcher cross-validated the qualitative interpretations generated by the LLM against the original rubric-based data, confirming that the interpretations accurately met performance criteria without semantic distortion or error. Through this process, the study was able to holistically diagnose the functional constraints and contextual causes of task failure for each cluster, ultimately yielding behavior-based diagnostic outcomes such as those illustrated in Figure 4-14. By integrating qualitative interpretation with quantitatively assessed performance data, the study identified specific support needs for each persona and derived concrete insights into the types and strategies of service interventions likely to be effective in their daily lives.

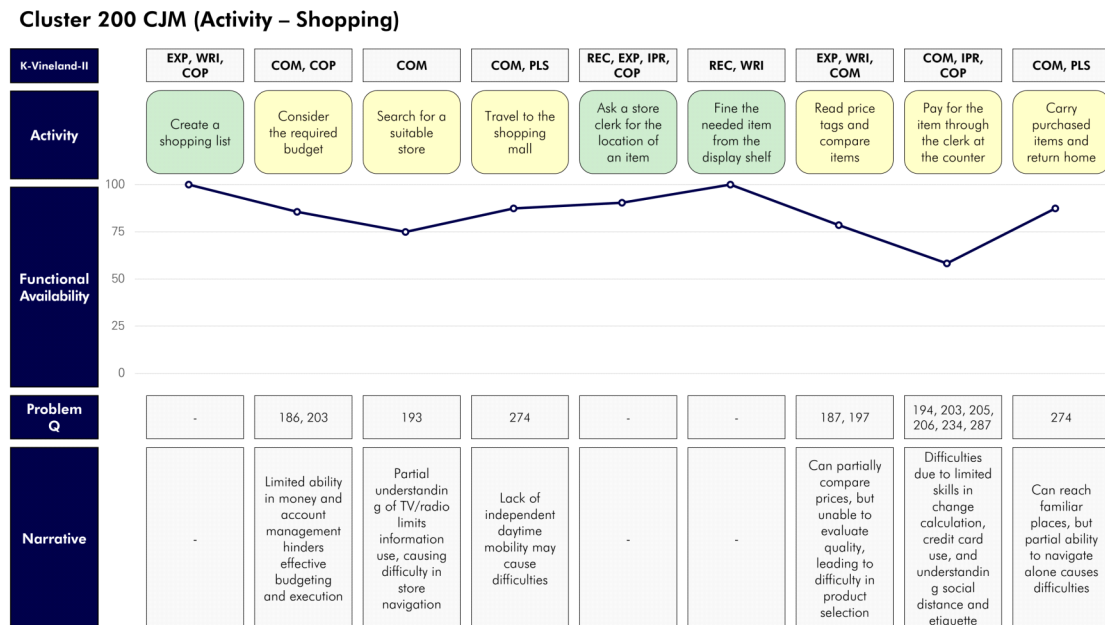


Figure 4-14. AI-driven Service Journey Map

4.3.3. Validation of Service Journey Map Fidelity

To verify the practical applicability of the service journey maps based on the representative personas derived in this study, the scenario performance data of two actual people with developmental disabilities—Cases G and H, which were excluded from the initial analysis—were compared with the journey map results of the representative personas corresponding to their respective clusters (Cluster 202 and Cluster 201). This comparison aimed to quantitatively assess the extent to which the representative personas reflect the actual performance characteristics of real users.

Three metrics were used for similarity analysis: cosine similarity, exact match

rate, and MAE similarity. Cosine similarity evaluates the overall directionality and pattern resemblance of journey map performance, providing an assessment of agreement at a macro level. Exact match rate measures the degree of alignment based on the performance of specific activities within each scenario. By distinguishing between full performance (2) and partial performance (1), this metric allows for the identification of specific areas where discrepancies exist between the actual user and the representative persona. In contrast, MAE similarity interprets partial performance (1) not as outright failure but as a matter of degree, thereby reflecting the continuity of performance levels and offering a more comprehensive measure of similarity.

Table 4-7. Validation of Service Journey Mapping for Case ‘G’

Domain	Scenarios		Cosine Similarity	Exact match rate	MAE Similarity
Personal	Self-Care	Toileting	0.983	0.844	0.891
		Toothbrushing	1.000	1.000	1.000
		Face washing	1.000	1.000	1.000
		Showering	1.000	1.000	1.000
		Dressing	0.989	0.860	0.908
Domestic	Meal		0.982	0.809	0.867
	Household Tasks		0.990	0.833	0.892
Community	Mobility		0.986	0.953	0.947
	Community		0.936	0.761	0.789
	Activity	Shopping	0.941	0.806	0.830
		Hospital Visit	0.955	0.816	0.838

Table 4-8. Validation of Service Journey Mapping for Case ‘H’

Domain	Scenarios		Cosine Similarity	Exact match rate	MAE Similarity
Personal	Self-Care	Toileting	1.000	1.000	1.000
		Toothbrushing	1.000	1.000	1.000
		Face washing	1.000	1.000	1.000
		Showering	1.000	1.000	1.000
		Dressing	1.000	0.929	0.964
Domestic	Meal		0.965	0.769	0.840
	Household Tasks		0.873	0.782	0.771
Community	Mobility		0.922	0.745	0.778
	Community		0.861	0.708	0.744
	Activity	Shopping	0.890	0.720	0.739
		Hospital Visit	0.984	0.947	0.944

The validation results showed that both G and H demonstrated very high cosine similarity scores ranging from 0.98 to 1.0 in the Self-Care domain, indicating a strong alignment in overall performance patterns with their respective representative personas. Especially for routine daily activities such as toothbrushing, face washing, and showering, not only the cosine similarity but also the exact match rate and MAE similarity were all 1.0, signifying complete performance similarity. In contrast, for scenarios with more variability or environmental factors—such as Meal, Household tasks, Mobility, and Community—the similarity scores slightly decreased but still maintained high concordance, with most values above 0.86. On average, the scores were: cosine similarity at 0.96, exact match rate at 0.86, and MAE similarity at 0.89. This suggests that the journey maps based on representative personas effectively reflect the actual performance characteristics of individuals. Minor discrepancies in similarity were due to differences in performance at specific activity levels. For example, in the “toileting” scenario within Self-Care, G showed an exact match rate of 0.844.

This was due to partial performance on certain tasks such as “undoing/fastening pants,” “move to the sink and turn on the water,” “wash hands,” “apply soap to hands,” and “rinsing hands.” These discrepancies were more flexibly reflected in the MAE similarity, which increased to 0.891 for the same scenario. This indicates that a continuous spectrum perspective, rather than binary evaluation, is more appropriate when assessing the performance of people with developmental disabilities.

Further analysis of performance alignment at the detailed activity level is presented in the appendix. While most scenarios showed strong similarity, some meaningful differences were identified. For instance, in G’s case, the exact match rate for “pa for the item through the clerk at the counter” was only 0.25, and similarly low scores were seen for “travel to the medical facility” (0.5) and “receive a prescription and complete payment” (0.25). These repeatedly observed discrepancies suggest that certain activities may deviate from the patterns represented by the persona, which naturally reflects the spectrum and individual variability inherent in developmental disabilities.

Such findings provide empirical support that the cluster-based representative persona methodology can effectively describe and predict the real-life behavioral performance of people with developmental disabilities. Notably, in basic self-sufficiency domains like Self-Care, the personas achieved over 90% alignment, indicating their potential as a valuable reference in early support planning. Moreover, by analyzing differences across the three similarity metrics, it becomes possible to systematically identify low-concordance areas, enabling targeted additional assessments and customized support planning without requiring full-scale evaluation.

In conclusion, this multi-metric validation confirms that journey maps based on representative personas not only accurately reflect the actual performance traits of people with developmental disabilities but also provide practical information necessary for individualized support planning. This demonstrates that the persona-based service design approach, grounded in scientific evidence, is a valid methodology capable of enhancing both the precision and efficiency of support systems for people with developmental disabilities.

Chapter 5. General Discussion

This study proposes a data-driven framework for constructing representative personas and structurally analyzing their service needs within daily living contexts, in order to overcome the limitations of previous research that struggled to apply service design to developmental disabilities due to their spectrum-like variability. By algorithmically augmenting a small number of disability profiles and validating their representativeness, the study addresses structural limitations in sample collection and expands the potential for data-driven analysis in disability research.

The proposed framework consists of the following components. First, a small dataset of developmental disability interviews was algorithmically augmented, and representative personas were derived through hierarchical clustering. Specifically, the original data from six individuals with developmental disabilities were expanded to a dataset of 2,406 profiles, which were then compressed into five representative personas. A comparative analysis was then conducted using data from two actual people with developmental disabilities who had been excluded from the initial analysis. The results confirmed that the generated personas successfully encompassed the functional range of real users, thereby validating their representativeness. This demonstrates the potential to overcome the sample limitations and subjectivity inherent in conventional qualitative persona construction methods in the disability domain, enabling repeatable and structured user modeling.

Second, the derived representative personas were applied to various daily living scenarios, where the level of service need was quantitatively assessed at the activity level and visualized in the form of a journey map. Notably, by employing large language models to explain the reasons for activity limitations, the study enabled a more systematic and objective identification of support needs—previously reliant on caregivers’ intuition and experiential observation. Furthermore, by comparing the journey map performance data of the representative personas with that of two real individuals, the study validated both the representativeness and practical utility of the proposed journey map-based evaluation method.

This study offers two key academic contributions. First, it establishes a quantitative persona modeling methodology for people with developmental disabilities based on the K-Vineland-II framework, thereby presenting the potential for interdisciplinary integration between traditional disability research and service design. Second, by quantifying the specific needs within the domain of daily living support services—arguably the most immediate area of support for people with developmental disabilities—the study advances the level of analysis from a general “identification of service need” to the more concrete “proposition of actionable intervention units.”

From an industrial perspective, the study holds the following implications. First, it enables practical service design that reflects the diverse life contexts of people with developmental disabilities, who exhibit strong spectrum variability. This, in turn, provides substantial practical value by significantly reducing the time and economic costs associated with persona development and service demand analysis. Second, the data augmentation-based framework proposed in

this study is not limited to people with developmental disabilities but is also applicable to other minority groups such as older adults and ethnic minorities, and even extends to general non-disabled populations. Third, by introducing persona-based user simulations, the study opens new avenues for experimental user research, enabling applications in UX design, emotional response evaluation, and market response forecasting. This approach can be adapted for design assessment across diverse variables such as products and services, usage contexts, and interaction types. Furthermore, it provides a quantitative foundation for comparing and analyzing the service experiences of disabled and non-disabled users, thereby offering empirical evidence to support the mainstreaming and strategic expansion of disability-related markets in the future.

However, this study has the following limitations. First, the original data used as the basis for data augmentation were limited in number and generally belonged to relatively high-functioning individuals, which led to a restricted range of differentiation in the clustering results. This was due to the algorithm being designed to preserve the uniqueness of the original data by limiting the range of variation. If more diverse data across varying levels of disability are secured in the future, the functional diversity and extrapolative potential of the representative personas can be further enhanced. Notably, the fact that one of the five representative personas derived in this study does not correspond to any of the original data points indicates this potential for scalability. Second, in some subdomains of functional ability, the match rate between the representative personas and actual users dropped to as low as 30%. This suggests that the currently constructed personas do not fully capture the entire functional

spectrum of people with developmental disabilities. Additionally, since the data in this study were collected using the structured assessment tool K-Vineland-II, the analysis was limited to the domains of adaptive behavior defined by the tool. In reality, however, the lives of people with developmental disabilities are more multidimensional, encompassing environmental contexts, subjective experiences, and digital interactions. Yet this study's analytical framework was constrained in its ability to fully reflect such complexity. Accordingly, in line with the first limitation, future research should secure more heterogeneous cases and utilize diverse assessment tools in a cross-referenced manner, allowing for deeper understanding of complex contexts and construction of more representative personas. Third, the application of LLMs was confined to linguistic interpretation of service needs, which also presents a limitation. In this study, the use of LLMs was limited to describing the reasons for functional limitations in each activity. Future studies should expand LLM applications to behavior prediction through simulation, evaluation of dynamic scenarios, and generation of interactive responses. By doing so, the full potential of LLMs can be more actively harnessed, thereby enhancing the precision and real-world relevance of data- and AI-driven service design.

This study has the potential to be expanded through the integration of LLM-based behavioral simulations and digital twin technologies, enabling applications such as real-time assessment, policy design, and personalized service development. In particular, by constructing composite personas that reflect not only functional capabilities but also individual traits such as personality and life patterns, it becomes possible to model people with disabilities in a more precise and nuanced manner. These advanced personas can

serve as experimental tools for simulating real-world service contexts within virtual environments, utilizing LLM-generated interview data and digital twin simulations embedded in metaverse platforms to predict user responses. Beyond conventional service design, such an approach offers the potential to simulate therapeutic responses in people with developmental disabilities and functional changes following medical interventions. This contributes to data-driven decision-making in clinical contexts and the formulation of personalized treatment strategies. Moreover, the proposed framework demonstrates scalability, offering a strategic design system applicable not only to people with disabilities and other minority groups, but also to general user populations.

In sum, this study empirically validates the feasibility of a data-driven approach to service design for people with developmental disabilities and proposes a novel direction for persona-based integrative design. By reframing disability-conscious design from a perspective of “specialization” to one of “integration,” the study aims to contribute to a paradigm shift in the field of strategic design for disability inclusion.

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Appendix

<Appendix 1> Validation of Service Journey Mapping in Detail

Table 4-9. Validation of Service Journey Mapping for Case ‘G’ (Detail)

Do ma in	Scenario		Activity	Cosine Similarity	Exact match rate	MAE Similarity
P e r s o n a l	Self- Care	Toileting	Move to the restroom	1.000	1.000	1.000
			Prepare the toilet	1.000	1.000	1.000
			Unfasten pants button	1.000	1.000	1.000
			Unzip pants	1.000	1.000	1.000
			Lower pants and underwear	1.000	1.000	1.000
			Sit on the toilet and void	1.000	1.000	1.000
			Wipe with toilet paper	1.000	1.000	1.000
			Flush the toilet	1.000	1.000	1.000
			Pull up pants and underwear	1.000	1.000	1.000
			Zip up pants	1.000	1.000	1.000
			Fasten pants button	1.000	1.000	1.000
			Move to the sink and turn on the water	1.000	1.000	1.000
			Wash hands	1.000	1.000	1.000
			Apply soap to hands	1.000	1.000	1.000
			Rinse hands	1.000	1.000	1.000
			Dry hands with a towel	1.000	1.000	1.000
		Tooth brushing	Pick up a toothbrush and toothpaste	1.000	1.000	1.000
			Apply toothpaste to the toothbrush	1.000	1.000	1.000
			Brush teeth with the toothbrush	1.000	1.000	1.000

			Brush the tongue with the toothbrush	1.000	1.000	1.000
			Rinse the mouth	1.000	1.000	1.000
			Rinse the toothbrush	1.000	1.000	1.000
			Wipe the mouth area with a towel	1.000	1.000	1.000
		Face washing	Dispense facial cleanser onto the hands	1.000	1.000	1.000
			Wash the face	1.000	1.000	1.000
			Rinse the face	1.000	1.000	1.000
			Dry the face with a towel	1.000	1.000	1.000
		Showering	Turn on the faucet and adjust water to an appropriate temperature	1.000	1.000	1.000
			Wet the hair and body	1.000	1.000	1.000
			Dispense shampoo	1.000	1.000	1.000
			Apply shampoo to the hair	1.000	1.000	1.000
			Rinse the hair thoroughly	1.000	1.000	1.000
			Dispense facial cleanser onto the hands	1.000	1.000	1.000
			Wash the face	1.000	1.000	1.000
			Rinse the face	1.000	1.000	1.000
			Dispense body wash	1.000	1.000	1.000
			Apply body wash to the body	1.000	1.000	1.000
			Rinse the body thoroughly	1.000	1.000	1.000
			Turn off the water	1.000	1.000	1.000
			Dry the hair and body with a towel	1.000	1.000	1.000
			Dry the hair using a hair dryer	1.000	1.000	1.000
		Dressing	Check the weather and daily schedule	1.000	1.000	1.000
			Select appropriate clothing based on the weather and context	1.000	0.000	0.500
			Put on underwear	1.000	1.000	1.000

D o m e s t i c			Put on pants	1.000	1.000	1.000
			Zip up pants	1.000	1.000	1.000
			Fasten pants button	1.000	1.000	1.000
			Put on a T-shirt	1.000	1.000	1.000
			Put on a shirt	1.000	1.000	1.000
			Button up the shirt	1.000	1.000	1.000
			Put on socks	1.000	1.000	1.000
			Put on a jacket	1.000	1.000	1.000
			Zip up the jacket	1.000	1.000	1.000
			Put on shoes	1.000	1.000	1.000
			Tie shoelaces	1.000	1.000	1.000
	Meal		Review the recipe	0.915	0.667	0.692
			Take out ingredients according to the recipe	0.816	0.000	0.429
			Wash the ingredients	1.000	1.000	1.000
			Chop and prepare the ingredients	0.962	0.667	0.750
			Heat the soup	0.956	0.600	0.714
			Stir-fry rice	0.949	0.500	0.667
			Take out side dishes	1.000	1.000	1.000
			Use the microwave to heat side dishes	1.000	1.000	1.000
			Set the table	1.000	1.000	1.000
			Eat the meal	1.000	1.000	1.000
			Take medication	0.976	0.800	0.833
			Clear the table	1.000	1.000	1.000
	Household tasks		Organize personal belongings in the room	1.000	1.000	1.000
			Vacuum the personal room	1.000	1.000	1.000
			Vacuum the living room	0.866	0.750	0.667
			Pre-rinse dishes with water	1.000	1.000	1.000
			Apply dish soap to the dishes	1.000	1.000	1.000
			Rinse off dishes with clean water	1.000	1.000	1.000
			Add detergent and fabric softener to the washing machine	1.000	1.000	1.000
			Load laundry into the washing machine and	0.980	0.833	0.857

C o m m u n i t y		start the cycle			
		Transfer washed clothes to the dryer and start the cycle	0.980	0.833	0.857
		Fold the clothes	1.000	1.000	1.000
		Organize the folded clothes	1.000	1.000	1.000
	Mobility	Check transportation information to the destination	0.913	0.833	0.750
		Walk to the bus stop	0.707	0.500	0.500
		Cross the crosswalk	0.816	0.667	0.600
		Wait at the bus stop	1.000	1.000	1.000
		Identify the correct bus to the destination	0.816	0.667	0.600
		Tap the transit card and board the bus	1.000	1.000	1.000
		Recognize arrival at the transfer stop	0.989	0.909	0.917
		Tap the transit card and get off the bus	1.000	1.000	1.000
		Recognize that one got off at the wrong stop	0.333	0.200	0.385
		Call a guardian to ask for assistance	0.892	0.714	0.700
		Use a smartphone to find an alternative route	0.816	0.667	0.600
		Identify an alternative arriving bus based on the new route	0.816	0.667	0.600
		Tap the transit card and board the bus	1.000	1.000	1.000
		Recognize arrival at the destination	0.989	0.909	0.917
		Tap the transit card and get off the bus	1.000	1.000	1.000
	School participation	Greet peers	1.000	1.000	1.000
		Transition to the next class according to the schedule	0.943	0.667	0.750
		Sit down at one's seat at	0.913	0.667	0.750

		the beginning of class			
		Respond when called during roll call	0.980	0.833	0.857
		Pay attention during class	0.714	0.143	0.438
		Read aloud or silently from the textbook	0.986	0.889	0.900
		Take notes on the class material	0.858	0.571	0.609
		Answer a question when called on by the teacher	0.961	0.850	0.833
		Discuss group activity topics with classmates	0.857	0.684	0.679
		Share personal belongings (e.g., pencil, eraser, toy) with peers	0.919	0.750	0.727
		Borrow an item from a peer	0.707	0.500	0.500
		Talk with friends during break time	0.869	0.630	0.659
		Express gratitude when receiving a gift from a peer	1.000	1.000	1.000
		Complete a task or errand assigned by the teacher	0.974	0.778	0.818
		Ask a peer for help	0.966	0.750	0.800
		Participate in basic individual physical activities (e.g., kicking a ball, vaulting)	1.000	1.000	1.000
		Participate in team sports (e.g., soccer, volleyball, basketball)	0.941	0.727	0.786
		Record homework or announcements during end-of-day homeroom	0.980	0.833	0.857

		Say goodbye to friends and leave school	0.943	0.889	0.818
	Activity – Shopping	Create a shopping list	0.976	0.833	0.857
		Consider the required budget	0.873	0.571	0.636
		Search for a suitable store	1.000	1.000	1.000
		Travel to the shopping mall	0.707	0.500	0.500
		Ask a store clerk for the location of an item	0.921	0.714	0.724
		Find the needed item from the display shelf	1.000	1.000	1.000
		Read price tags and compare items before making a selection	1.000	1.000	1.000
		Pay for the item through the clerk at the counter	0.564	0.250	0.480
		Carry purchased items and return home	0.707	0.500	0.500
	Activity – Hospital visit	Recognize symptoms such as cold, fever, chest pain, or injury	1.000	1.000	1.000
		Take available medication and recognize if medication is insufficient	0.962	0.667	0.750
		Search for a nearby clinic or hospital	0.894	0.800	0.714
		Inform a guardian and request assistance	0.929	0.824	0.773
		Call to make an appointment at a hospital or clinic	0.971	0.750	0.800
		Travel to the medical facility	0.707	0.500	0.500
		Describe symptoms at the reception desk	0.992	0.933	0.938
		Explain symptoms during	0.993	0.941	0.944

		a consultation with the doctor			
		Receive a prescription and complete payment	0.634	0.200	0.476
		Go to the pharmacy	0.866	0.750	0.667
		Present the prescription to the pharmacist	0.946	0.786	0.778
		Receive the medication and complete payment	0.889	0.625	0.667
		Return home	0.707	0.500	0.500
		Take the prescribed medication	0.976	0.800	0.833

Table 4-10. Validation of Service Journey Mapping for Case 'H' (Detail)

Do ma in	Scenario		Activity	Cosine Similarity	Exact match rate	MAE Similarity
P e r s o n a l	Self-Care	Toileting	Move to the restroom	1.000	1.000	1.000
			Prepare the toilet	1.000	1.000	1.000
			Unfasten pants button	0.962	0.667	0.750
			Unzip pants	1.000	1.000	1.000
			Lower pants and underwear	1.000	1.000	1.000
			Sit on the toilet and void	1.000	1.000	1.000
			Wipe with toilet paper	1.000	1.000	1.000
			Flush the toilet	1.000	1.000	1.000
			Pull up pants and underwear	1.000	1.000	1.000
			Zip up pants	1.000	1.000	1.000
			Fasten pants button	0.962	0.667	0.750
			Move to the sink and turn on the water	0.962	0.667	0.750
			Wash hands	0.949	0.500	0.667
			Apply soap to hands	0.949	0.500	0.667
			Rinse hands	0.949	0.500	0.667
			Dry hands with a towel	1.000	1.000	1.000
		Tooth brushing	Pick up a toothbrush and toothpaste	1.000	1.000	1.000
			Apply toothpaste to the toothbrush	1.000	1.000	1.000
			Brush teeth with the toothbrush	1.000	1.000	1.000
			Brush the tongue with the toothbrush	1.000	1.000	1.000
			Rinse the mouth	1.000	1.000	1.000
			Rinse the toothbrush	1.000	1.000	1.000
			Wipe the mouth area with a towel	1.000	1.000	1.000
		Face washing	Dispense facial cleanser onto the hands	1.000	1.000	1.000
			Wash the face	1.000	1.000	1.000
			Rince the face	1.000	1.000	1.000
			Dery the face with a	1.000	1.000	1.000

			towel			
		Showering	Turn on the faucet and adjust water to an appropriate temperature	1.000	1.000	1.000
			Wet the hair and body	1.000	1.000	1.000
			Dispense shampoo	1.000	1.000	1.000
			Apply shampoo to the hair	1.000	1.000	1.000
			Rinse the hair thoroughly	1.000	1.000	1.000
			Dispense facial cleanser onto the hands	1.000	1.000	1.000
			Wash the face	1.000	1.000	1.000
			Rinse the face	1.000	1.000	1.000
			Dispense body wash	1.000	1.000	1.000
			Apply body wash to the body	1.000	1.000	1.000
			Rinse the body thoroughly	1.000	1.000	1.000
			Turn off the water	1.000	1.000	1.000
			Dry the hair and body with a towel	1.000	1.000	1.000
			Dry the hair using a hair dryer	1.000	1.000	1.000
		Dressing	Check the weather and daily schedule	0.935	0.875	0.800
			Select appropriate clothing based on the weather and context	1.000	0.000	0.500
			Put on underwear	1.000	1.000	1.000
			Put on pants	1.000	1.000	1.000
			Zip up pants	1.000	1.000	1.000
			Fasten pants button	0.949	0.500	0.667
			Put on a T-shirt	1.000	1.000	1.000
			Put on a shirt	1.000	1.000	1.000
			Button up the shirt	0.962	0.667	0.750
			Put on socks	1.000	1.000	1.000
			Put on a jacket	1.000	1.000	1.000
			Zip up the jacket	1.000	1.000	1.000
			Put on shoes	1.000	1.000	1.000
			Tie shoelaces	1.000	1.000	1.000

D o m e s t i c	Meal	Review the recipe	0.971	0.778	0.818
		Take out ingredients according to the recipe	1.000	0.333	0.600
		Wash the ingredients	1.000	1.000	1.000
		Chop and prepare the ingredients	0.962	0.667	0.750
		Heat the soup	0.941	0.600	0.714
		Stir-fry rice	0.926	0.500	0.667
		Take out side dishes	1.000	1.000	1.000
		Use the microwave to heat side dishes	1.000	1.000	1.000
		Set the table	1.000	1.000	1.000
		Eat the meal	1.000	1.000	1.000
		Take medication	1.000	1.000	1.000
		Clear the table	0.980	0.833	0.857
	Household tasks	Organize personal belongings in the room	1.000	1.000	1.000
		Vacuum the personal room	0.971	0.750	0.800
		Vacuum the living room	0.965	0.750	0.800
		Pre-rinse dishes with water	1.000	1.000	1.000
		Apply dish soap to the dishes	1.000	1.000	1.000
		Rinse off dishes with clean water	1.000	1.000	1.000
		Add detergent and fabric softener to the washing machine	1.000	0.000	0.500
		Load laundry into the washing machine and start the cycle	0.980	0.833	0.857
		Transfer washed clothes to the dryer and start the cycle	0.980	0.833	0.857
		Fold the clothes	1.000	1.000	1.000
		Organize the folded clothes	1.000	1.000	1.000
C o m	Mobility	Check transportation information to the destination	0.955	0.917	0.857

m u n i t y		Walk to the bus stop	1.000	1.000	1.000
		Cross the crosswalk	1.000	1.000	1.000
		Wait at the bus stop	1.000	1.000	1.000
		Identify the correct bus to the destination	1.000	1.000	1.000
		Tap the transit card and board the bus	1.000	1.000	1.000
		Recognize arrival at the transfer stop	0.989	0.909	0.917
		Tap the transit card and get off the bus	1.000	1.000	1.000
		Recognize that one got off at the wrong stop	0.970	0.800	0.833
		Call a guardian to ask for assistance	0.990	0.929	0.933
		Use a smartphone to find an alternative route	0.900	0.833	0.750
		Identify an alternative arriving bus based on the new route	1.000	1.000	1.000
		Tap the transit card and board the bus	1.000	1.000	1.000
		Recognize arrival at the destination	0.989	0.909	0.917
		Tap the transit card and get off the bus	1.000	1.000	1.000
	School participation	Greet peers	0.853	0.727	0.647
		Transition to the next class according to the schedule	0.962	0.667	0.750
		Sit down at one's seat at the beginning of class	0.775	0.333	0.500
		Respond when called during roll call	0.980	0.833	0.857
		Pay attention during class	0.918	0.429	0.636
		Read aloud or silently from the textbook	1.000	1.000	1.000
		Take notes on the class material	0.920	0.500	0.667

		Answer a question when called on by the teacher	0.979	0.850	0.870
		Discuss group activity topics with classmates	0.961	0.895	0.864
		Share personal belongings (e.g., pencil, eraser, toy) with peers	0.791	0.625	0.571
		Borrow an item from a peer	1.000	1.000	1.000
		Talk with friends during break time	0.953	0.778	0.794
		Express gratitude when receiving a gift from a peer	1.000	1.000	1.000
		Complete a task or errand assigned by the teacher	0.974	0.778	0.818
		Ask a peer for help	0.935	0.875	0.800
		Participate in basic individual physical activities (e.g., kicking a ball, vaulting)	1.000	1.000	1.000
		Participate in team sports (e.g., soccer, volleyball, basketball)	0.924	0.636	0.733
		Record homework or announcements during end-of-day homeroom	0.969	0.750	0.800
		Say goodbye to friends and leave school	0.882	0.778	0.692
	Activity – Shopping	Create a shopping list	0.944	0.667	0.750
		Consider the required budget	0.964	0.714	0.778
		Search for a suitable store	0.707	0.500	0.500
		Travel to the shopping mall	1.000	1.000	1.000
		Ask a store clerk for the	0.988	0.905	0.913

		location of an item			
		Find the needed item from the display shelf	1.000	1.000	1.000
		Read price tags and compare items before making a selection	0.957	0.714	0.778
		Pay for the item through the clerk at the counter	0.908	0.750	0.750
		Carry purchased items and return home	1.000	1.000	1.000
Activity - Hospital visit		Recognize symptoms such as cold, fever, chest pain, or injury	0.949	0.500	0.667
		Take available medication and recognize if medication is insufficient	1.000	1.000	1.000
		Search for a nearby clinic or hospital	0.874	0.800	0.714
		Inform a guardian and request assistance	0.986	0.882	0.895
		Call to make an appointment at a hospital or clinic	0.923	0.500	0.667
		Travel to the medical facility	1.000	1.000	1.000
		Describe symptoms at the reception desk	0.982	0.867	0.882
		Explain symptoms during a consultation with the doctor	0.976	0.824	0.850
		Receive a prescription and complete payment	0.905	0.700	0.714
		Go to the pharmacy	0.928	0.875	0.800
		Present the prescription to the pharmacist	0.980	0.857	0.875
		Receive the medication and complete payment	0.870	0.625	0.667
		Return home	1.000	1.000	1.000

		Take the prescribed medication	1.000	1.000	1.000
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논문 요약

데이터 기반 장애인 페르소나 증폭 및 서비스 여정 추론 연구: 발달장애인의 일상생활지원수요를 중심으로

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장애인을 위한 서비스디자인은 그 필요성과 잠재적 가치에도 불구하고, 소수성과 시장성 부족이라는 이중의 제약으로 인해 학문적·실무적 관심에서 상대적으로 저평가되어 왔다. 특히 발달장애인의 경우, 스펙트럼성으로 인해 개인 간 차이가 극단적으로 다양하게 분포함에 따라, 일반화와 대표성 확보가 어려워 설계 및 평가 과정에서 정량적 분석이 난해하다. 본 연구는 이와 같은 발달장애인의 소수성, 스펙트럼성, 산발성이라는 구조적 문제를 해결하기 위한 대안으로, 데이터 증폭 기법과 알고리즘적 클러스터링을 활용하여 대표 페르소나를 도출하고, 이를 일상 시나리오에 매핑하여 일상생활지원서비스 니즈를 정량적으로 진단하는 Augmented Persona Mapping Framework를 제안한다.

본 연구는 두 가지 목표를 중심으로 설계되었다. 첫 번째 목표는 제한된 수의 실제 발달장애인 데이터를 증폭하여 스펙트럼 다양성을 확보함으로써, 일반화 가능한 대표 페르소나를 정의할 수 있는 알고리즘을 개발하는 것이다. 두 번째 목표는 도출된 대표 페르소나를 일상생활지원 시나리오에 대입하여, 활동 단위별로 필요한

기능과 제약 요인을 구조화하고, 서비스 니즈 수준을 수치화하는 정량적 분석 체계를 마련하는 것이다.

이를 위해 다음과 같은 연구가 진행되었다. 우선, 실제 발달장애인 6명의 인터뷰 데이터를 K-Vineland-II 알고리즘에 기반하여 총 2,406명의 가상 데이터로 증폭하였다. 이후 주성분 분석 및 계층적 클러스터링을 통해 기능 특성에 따라 5개의 대표 페르소나를 도출하였다. 이후 대표 페르소나를 일상생활 기반 시나리오(e.g., 학교 생활, 교통 이용, 식사 준비, 쇼핑 등)에 대입하여, 각 시나리오 수행을 위한 활동 단위별로 필요한 기능 및 적응행동 요소에 대한 수행 제약 요인을 대형 언어 모델을 통해 해석하였다. 수치화하여 분석된 서비스 니즈 수준 및 세부 내용은 고객 여정 지도 형태로 시각화하였다.

본 프레임워크의 유효성은 실제 발달장애인 사례의 수행 데이터를 기준으로 검증하였으며, 그 결과 대표 페르소나는 실제 발달장애인의 장애 수준에 대해 75% 이상의, 이에 기반한 저니맵은 실제 발달장애인의 일상 수행 경향과 86% 이상 수준의 높은 정합도를 나타냈다. 이를 통해 해당 프레임워크가 활동 제한의 원인과 서비스 개입 필요 수준을 구체적으로 제시할 수 있는 분석 도구로 기능함을 확인하였다. 이는 장애인의 일상생활 내 서비스 니즈를 직관이 아닌 구조적·반복 가능한 방식으로 해석할 수 있음을 시사한다.

본 연구는 장애인의 행동 데이터를 구조화·증폭하여 페르소나 및 저니맵 설계에 활용할 수 있는 가능성을 실증함으로써, 정성 중심의 기존 장애 연구에 정량 기반 접근을 도입하고, 실증성과 반복 가능성을 동시에 확보하는 분석 프레임워크를 제시하였다는 점에서 의의가 있다. 향후에는 이를 디지털 트윈 기반 UX 설계 및 AI 기반 서비스디자인과 연계함으로써, 사용자 반응 예측 및 시뮬레이션 기반 서비스 전략 수립으로 확장할 수 있을 것으로 기대된다.

주제어 : 데이터드리븐, 서비스디자인, 발달장애, 증폭, 일상생활지원서비스